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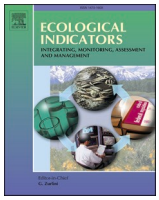

Experiential factors associated with subjective well-being in urban waterfront environments: Evidence from social media data in Wuhan, China

Ke Wang [a], Xia Zhang [a,*], Hong Yang [b], Luliang Tang [b], Dong Xie [a], Cunyu Yuan [a]

[a] School of Urban Design, Wuhan University, Wuhan, PR China
[b] State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, PR China

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ABSTRACT

Urban waterfront environments, as important forms of urban blue spaces, are increasingly recognized for their contributions to human health and subjective well-being (SWB). However, previous studies have mainly classified waterfront spaces based on waterbody attributes and examined the relationships between objectively measured environmental characteristics and health outcomes, while how spatial forms and functional attributes are associated with public experiences and how these experiential differences relate to SWB remain underexplored. To address these gaps, this study utilizes large-scale social media data from Wuhan to explore how experiential factors—including perception, cognition, and behavior—are associated with SWB in urban waterfront environments and examines their temporal dynamics and typological disparities. The results reveal pronounced long-term variations and seasonal cycles in SWB across waterfront environments that are associated with social and climatic conditions. Overall, physical activity shows the strongest association with SWB (19.12%), followed by traffic (15.96%) and crowding (9.89%). Distinct patterns emerge across waterfront typologies: riverfronts, characterized by a balanced mix of exercise and leisure, exhibit the highest overall SWB with moderate seasonality; waterfront parks, more strongly associated with accessibility and natural environmental quality, show slightly lower SWB but the greatest year-round stability; and waterfront greenways, characterized by physical activity, display the lowest levels of SWB with the largest seasonal variability. This study highlights general experiential patterns and typology-specific differences associated with SWB in waterfront environments. It provides empirical evidence for understanding the health-related benefits of urban blue spaces and offers practical insights for the design and management of waterfront environments to promote public well-being.

1. Introduction

Urbanization has generated substantial economic and social benefits, yet it has also intensified environmental pressures and health challenges (Flies et al., 2019; Lorenc et al., 2012). This has spurred growing interest in how urban environments can be planned to support population health and advance sustainable development agendas (Adlakha and John, 2022; Hua et al., 2022). Among various types of urban public spaces, waterfront environments play a particularly important role. In addition to regulating microclimates and improving ecological conditions, they provide valuable settings for recreation, physical activity, and social interaction, thereby demonstrating considerable potential to promote human health and well-being (Gascon et al., 2017; Vian et al., 2021). Evidence shows that residents living near waterfronts tend to report better overall health, lower psychological stress, and greater life satisfaction and happiness (Wheeler et al., 2012, 2015; White et al., 2013), alongside more opportunities for physical exercise and social participation (Pasanen et al., 2019; Ashbullby et al., 2013). Building upon existing models of the health benefits of green spaces, White et al. proposed that blue spaces can improve individual health and well-being through multiple pathways, including reducing environmental stress, strengthening connections with nature, and facilitating attention restoration (White et al., 2020). Despite this growing recognition, research on the health and well-being benefits of waterfronts remains comparatively

* Corresponding author at: Wuhan University, Bayi Road 299, Wuhan, Hubei Province, PR China.
E-mail address: xiazhang@whu.edu.cn (X. Zhang).

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城市滨水环境中与主观幸福感相关的体验性因素：基于中国武汉社交媒体数据的实证研究

王凯[a]、张霞[a,*]、杨洪[b]、唐璐亮[b]、谢东[a]、袁存玉[a] [a]武汉大学城市设计学院，中国武汉

[b] 武汉大学测绘与遥感信息工程国家重点实验室，中国武汉

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ABSTRACT

城市滨水环境作为城市蓝色空间的重要形式，因其对人类健康及主观幸福感（SWB）的贡献而日益受到重视。然而，既往研究主要根据水体特征对滨水空间进行分类，并探讨客观测量的环境特征与健康结局之间的关联，而空间形态与功能属性如何与公众体验相关联，以及这些体验差异又如何与 SWB 相关联，则仍未得到充分研究。为填补这些空白，本研究利用武汉的大规模社交媒体数据，探究城市滨水环境中体验因素——包括感知、认知与行为——与 SWB 的关联，并分析其时间动态及类型学差异。研究结果显示，不同滨水环境中的 SWB 存在显著的长期变化及季节性波动，这些变化与社会及气候条件密切相关。总体而言，体力活动与 SWB 的关联最为显著（19.12%），其次是交通状况（15.96%）与拥挤程度（9.89%）。不同滨水类型中呈现出截然不同的模式：以运动与休闲均衡结合为特征的河岸区，整体 SWB 值最高，且季节性波动适中；与可达性及自然环境质量关联更为紧密的滨水公园， SWB 值略低，但全年稳定性最高；而以体力活动为特征的滨水绿道， SWB 值最低，且季节性波动最为显著。本研究揭示了滨水环境中与 SWB 相关的普遍体验模式及特定类型间的差异，为理解城市蓝色空间对健康的益处提供了实证依据，并为滨水环境的设计与管理提供了实用见解，以促进公众福祉。

1. 引言

城市化已带来巨大的经济与社会效益，但也加剧了环境压力与健康挑战（Flies 等，2019；Lorenc 等，2012）。这促使人们日益关注如何通过规划城市环境来促进居民健康并推进可持续发展议程（Adlakha 与 John，2022；Hua等，2022）。在众多城市公共空间类型中，滨水环境发挥着尤为重要的作用。除调节微气候与改善生态条件外，它们还为休闲活动、体育锻炼及社交互动提供了优质场所，从而展现出促进人类健康与福祉的巨大潜力（Gascon 等，2017；Vian 等，2021）。研究表明，居住于滨水区域的居民往往报告整体健康状况更佳、心理压力更低，且生活满意度与幸福感更高（Wheeler等，2012,2015；White等，2013），同时拥有更多体育锻炼与社交机会。参与性（Pasanen 等，2019；Ashbullby 等，2013）。基于现有关于绿地健康益处的研究模型，White 等人提出，蓝色空间可通过多种途径改善个人健康与福祉，包括减轻环境压力、加强与自然联系以及促进注意力恢复（White 等，2020）。尽管这一认识日益普及，但相较于关于城市绿地的广泛研究，关于滨水空间对健康与福祉益处的研究仍相对有限。这一研究空白阻碍了我们对滨

* 通讯作者：中国湖北省武汉市八一路299号，武汉大学。
*电子邮箱地址：*xiazhang@whu.edu.cn（X. Zhang）。

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limited relative to the extensive scholarship on urban green spaces. This gap hampers a fuller understanding of the associations between waterfront environments and human health and constrains their incorporation into health-oriented planning and policy practices (Grellier et al., 2017; Hunter et al., 2023). As a result, elucidating how waterfront spaces relate to residents' well-being and how to improve their environmental quality has become a key interdisciplinary concern across environmental psychology, public health, and urban planning.

Subjective well-being (SWB) is described as a psychological state that encompasses individuals' evaluative judgments about their lives and their affective responses to personal experiences (OECD, 2013). A deeper understanding of SWB holds fundamental significance, as it provides a scientific basis for formulating and evaluating broad social policies. Recent research consistently shows that waterfront spaces can substantially enhance individual well-being: more frequent visits, longer dwell times, and higher-intensity participation are all associated with higher levels of SWB (Vitale et al., 2022; White et al., 2021). Traditionally, these studies have relied on self-reported questionnaires and interviews that are costly and time-consuming, thereby limiting scalability for population-level applications (McDougall et al., 2022; Vitale et al., 2022; White et al., 2021). Conceptually, SWB is typically framed as comprising two interrelated dimensions: (i) cognitive evaluations of one's life as a whole, commonly termed life satisfaction; and (ii) experienced well-being, namely the stream of affective states (e.g., pleasure, tension) arising in everyday contexts (National Research Council, 2013). Studies increasingly use the intensity and frequency of momentary affective states as indicators of experienced well-being, assuming that affective responses directly reflect and quantify individual happiness in specific contexts (Y. Lin et al., 2022; Tu et al., 2026). Building on this perspective, the present study operationalizes SWB using sentiment-derived emotional expressions from social media data, focusing on its affective dimension.

It is noteworthy that the health-related effects of waterfront environments vary considerably depending on their types, scales, and functions. Existing studies have predominantly compared waterfronts based on waterbody types (Wang et al., 2024). For instance, McDougall et al. (2022) found that frequent visits to rivers and canals are associated with better mental health, whereas lake visits show no significant association. Luo et al. (2023) further revealed that during viewing and walking activities, rivers and lakes generally offer stronger restorative effects than canals. However, such classifications centered solely on waterbody type fail to capture the complexity and diversity of contemporary urban waterfronts in terms of their spatial morphology, functional attributes, and patterns of use. To address this limitation, recent research has begun to adopt more spatially grounded approaches, examining specific categories of waterfront spaces. For example, Zhang et al. (2024) examined Guangzhou's Pearl Riverfront, using street view segmentation, POI data, and heatmaps to quantify visual indices and explore drivers of waterfront vitality. Hu et al. (2026) focused on Shanghai's Huangpu River Greenway, applying seasonal wearable street view imagery and random forest models to reveal how built environments and seasonal microclimates shape use intensity. Li et al. (2025) investigated Shanghai's waterfront parks, using social media text mining and sentiment analysis to identify key cultural ecosystem services and differentiated drivers of user satisfaction. These studies, focusing respectively on riverfront, waterfront greenway, and waterfront park, reflect a broader academic shift from “waterbody-based classification” to “space-type classification,” offering new perspectives for understanding how different types of waterfront spaces are used and how they influence health outcomes.

The internal quality and evaluation of waterfront spaces are crucial for a comprehensive understanding of their relationships with health and well-being (Bell et al., 2021). Case studies from Plymouth in the United Kingdom and Barcelona in Spain have demonstrated that improvements in spatial quality can significantly enhance users' happiness and life satisfaction (Van Den Bogerd et al., 2021; Vert et al., 2019). In recent years, some scholars have begun to quantify the quality of waterfront spaces from an objective environmental perspective. For instance, Chen and Yuan (2020) quantified the fragmentation of blue spaces within a one-kilometer buffer and combined field surveys with street-view images to assess hydrophilic characteristics and analyze their impacts on human health. Garrett et al. (2023) proposed an evaluation framework encompassing indicators such as safety, cleanliness, facility integrity, and the presence of wildlife. Similarly, Cao et al. (2024) incorporated both visual elements—such as sky, water, vegetation, and buildings—and spatial factors—such as land use, building density, and the accessibility of transportation and service facilities—to examine their effects on public emotions.

In recent years, the proliferation of social media data has provided new opportunities to examine the interactions between humans and urban spaces (S. Chen et al., 2022a; Shen and Li, 2023). Social media platforms offer large-scale, wide-coverage, and cost-effective data sources that can capture individuals' natural expressions in real-world contexts (Y. Lin et al., 2022; X. Zhu et al., 2021). Meanwhile, advances in natural language processing (NLP) and machine learning have further enhanced the precision and interpretability of text mining, enabling researchers to extract experiential features, identify emotional tendencies, and uncover the complex relationships between them (Li et al., 2024). These developments provide a novel methodological approach to address the representativeness and timeliness limitations of traditional approaches and to explore the complex associations between waterfront environments and public happiness.

Despite increasing research attention, the relationship between urban waterfront environments and SWB remains insufficiently understood. First, existing studies have provided limited insights into the differences among various types of waterfront spaces. Previous research has commonly classified blue spaces based on waterbody attributes (e.g., rivers, lakes, coasts, and canals). Although this classification approach effectively captures the hydrological characteristics of waterfront environments, it may overlook the differences among urban waterfront spaces arising from variations in spatial morphology, functional positioning, and patterns of use. For example, waterfront greenways and waterfront parks may both be located along rivers, yet their distinct spatial organizations and functional characteristics can generate different behavioral patterns and experiential processes (Vian et al., 2021). How these spatial and functional differences are reflected in public experiences and further associated with SWB remains insufficiently explored. Therefore, this study classifies urban waterfront spaces into three typical types based on spatial forms and planning functions: linear waterfront greenways, belt-shaped riverfront spaces, and areal waterfront parks (Y. Chen et al., 2022b), and systematically compares the differences in experiential characteristics and SWB associations across these waterfront typologies (Table 1).

Second, existing studies on the health benefits of waterfront spaces have predominantly adopted an objective environmental perspective, with limited attention given to the experiential processes through which individuals interact with environments. Previous research has commonly used objective indicators, including environmental exposure, landscape characteristics, and spatial accessibility, to characterize waterfront environments and further examine their associations with health outcomes. These studies have provided valuable insights into identifying health-promoting environmental attributes; however, they often provide limited explanations of how individuals perceive, interpret, and interact with spatial environments. Environmental psychology theories suggest that the effects of environments on human health and well-being are not achieved through direct pathways alone, but rather through processes involving environmental perception, cognitive appraisal, and behavioral responses (Leslie et al., 2010; Luo et al., 2021). Compared with objective environmental attributes, individuals' subjective perceptions of environments are often more direct predictors of spatial use decisions and behavioral choices (Leslie et al., 2010), while behavioral activities also serve as an important pathway linking

水环境与人类健康之间关联的全面理解，也限制了将其纳入以健康为导向的规划与政策实践（Grellier等，2017；Hunter等，2023）。因此，阐明滨水空间与居民福祉的关联性及如何提升其环境质量，已成为环境心理学、公共卫生与城市规划领域跨学科研究的关键议题。

主观幸福感（SWB）被描述为一种心理状态，它涵盖了个体对其生活的评估判断以及对个人体验的情感反应（OECD，2013）。对 SWB 的更深入理解具有重要意义，因为它为制定和评估广泛的社会政策提供了科学依据。近期研究一致表明，滨水空间可显著提升个体的幸福感：更频繁的到访、更长的停留时间以及更高强度的参与，均与更高的 SWB 水平相关（Vitale 等，2022；White 等，2021）。传统上，这些研究依赖于自填式问卷和访谈，这些方法成本高昂且耗时，从而限制了其在人群层面应用的可扩展性（McDougall 等，2022；Vitale 等，2022；White 等，2021）。从概念上讲，SWB 通常被界定为包含两个相互关联的维度：(i) 对个人整体生活的认知评估，通常称为生活满意度；以及 (ii) 体验到的幸福感，即在日常情境中产生的情感状态流（如愉悦、紧张）（美国国家研究委员会，2013）。越来越多的研究将瞬时情感状态的强度与频率作为体验到的幸福感的指标，假设情感反应可直接反映并量化个体在特定情境下的幸福感（Y.Lin 等，2022；Tu 等，2026）。基于这一视角，本研究利用社交媒体数据中的情感衍生情绪表达，聚焦于 SWB 的情感维度，将其操作化。

值得注意的是，滨水环境对健康的影响因其类型、规模和功能的不同而存在显著差异。现有研究大多基于水体类型对滨水环境进行比较（Wang等，2024）。例如，McDougall等（2022）发现，频繁前往河流与运河区域与更好的心理健康状况相关，而前往湖泊区域则未显示出显著关联。Luo 等（2023）进一步指出，在观赏与步行活动中，河流与湖泊通常比运河提供更强的恢复效应。然而，此类仅基于水体类型的分类方式，未能充分捕捉当代城市滨水环境在空间形态、功能属性及使用模式方面的复杂性与多样性。为克服这一局限，近期研究已开始采用更具空间基础性的研究方法，针对特定类型的滨水空间展开分析。例如，Zhang等（2024）以广州珠江滨水区为例，运用街景分割、POI数据及热力图，量化视觉感知指标。探索滨水空间的活力驱动因素。Hu 等人（2026）以上海黄浦河绿道为研究对象，运用季节性可穿戴街景图像与随机森林模型，揭示建成环境与季节性微气候如何塑造空间使用强度。Li 等人（2025）则针对上海滨水公园展开研究，采用社交媒体文本挖掘与情感分析技术，识别关键文化生态系统服务及影响用户满意度的差异化驱动因素。这些分别聚焦河岸、滨水绿道与滨水公园的研究，体现了学术界从“水体导向分类”向“空间类型分类”的广泛转向，为理解不同类型滨水空间的利用方式及其对健康结局的影响提供了新的视角。

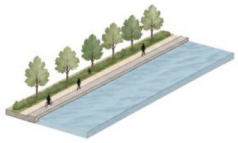
滨水空间的内在品质与评估，对于全面理解其与健康及福祉之间的关联至关重要（Bell 等，2021）。英国普利茅斯与西班牙巴塞罗那的案例研究表明，空间品质的提升可显著增强使用者的幸福感与生活满意度（Van Den Bogerd等，2021；Vert等，2019）。近年来，部分学者已开始从客观环境视角对滨水空间品质进行量化研究。例如，Chen 与 Yuan（2020）量化了一公里缓冲区内蓝色空间的碎片化程度，并将实地调查与街景图像相结合，以评估亲水特性及其对人类健康的影响。Garrett 等（2023）提出了一个评估框架，涵盖安全性、清洁度、设施完整性及野生动物存在等指标；同样，Cao 等（2024）则整合了视觉要素——如天空、水体、植被与建筑——以及空间要素——如土地利用等。城市密度、交通与服务设施的可达性——旨在探究这些因素对公众情绪的影响。

近年来，社交媒体数据的激增为探究人类与城市空间之间的互动提供了新的机遇（S.Chen 等，2022a；Shen 与 Li，2023）。社交媒体平台提供了大规模、广覆盖且成本效益高的数据源，能够捕捉个体在真实场景中的自然表达（Y.Lin 等，2022；X.Zhu 等，2021）。与此同时，自然语言处理（NLP）与机器学习技术的进步，进一步提升了文本挖掘的精准度与可解释性，使研究人员得以提取体验性特征、识别情感倾向，并揭示这些特征间的复杂关联（Li等，2024）。这些进展为应对传统研究方法在代表性与时效性方面的局限，以及探索滨水环境与公众幸福感之间的复杂关联，提供了一种全新的方法论路径。

尽管相关研究日益增多，城市滨水环境与 SWB 之间的关系仍尚未得到充分理解。首先，现有研究对不同类型滨水空间的差异探讨有限。既往研究通常根据水体属性（如河流、湖泊、海岸及运河）对蓝色空间进行分类。尽管这种分类方法能有效捕捉滨水环境的水文特征，却可能忽略因空间形态、功能定位及使用模式差异而导致的城市滨水空间之间的区别。例如，滨水绿道与滨水公园虽均坐落于河流沿岸，但其独特的空间布局与功能特性却可能催生不同的行为模式与体验过程（Vian 等，2021）。这些空间与功能差异如何在公众体验中体现，以及其与 SWB 的关联机制，目前仍缺乏充分探讨。因此，本研究根据空间形态与规划功能，将城市滨水空间划分为三种典型类型：线性滨水绿道、带状滨水空间及片状滨水公园（Y.Chen et al.，2022b），并系统比较了这些滨水空间类型在体验特征与 SWB 关联方面的差异（表1）。

其次，关于滨水空间健康效益的现有研究大多采用客观环境视角，而对个体与环境互动的体验过程关注有限。既往研究通常使用客观指标——包括环境暴露度、景观特征及空间可达性——来刻画滨水环境，进而探讨这些环境因素与健康结局之间的关联。这些研究为识别促进健康的环境属性提供了宝贵见解；然而，它们往往对个体如何感知、解读及与空间环境互动方面的解释较为有限。环境心理学理论指出，环境对人类健康与福祉的影响并非仅通过直接途径实现，而是经由环境感知、认知评估及行为反应等过程共同作用而成（Leslie 等，2010；Luo 等，2021）。与客观环境属性相比，个体对环境的主观感知往往更能直接预测空间使用决策与行为选择（Leslie 等，2010），而行为活动同样可作为连接环境条件与健康结局的重要途径（Liu 等，2019；Murrin 等，

Table 1
Data for 14 waterfront study sites.

Type	Site	Reviews (n)	Period	Spatial morphology
Waterfront greenway (8643)	a. East Lake Greenway	8643	2019/01/14–2024/12/30	 Linear
Riverfront (4936)	b. Hankou Jiangtan	1553	2019/02/06–2024/12/30	Belt
	c. Wuchang Jiangtan	1314	2019/01/12–2024/12/24	
	d. Qingshan Jiangtan	1185	2019/01/26–2024/12/23	
	e. Hanyang Jiangtan	884	2019/01/03–2024/12/26	
Waterfront park (6841)	f. Sha Lake Park	2356	2019/01/05–2024/12/27	Areal
	g. Ziyang Lake Park	1263	2019/02/08–2024/12/27	
	h. Moshui Lake Park	603	2019/01/05–2024/12/29	
	i. Moon Lake Park	556	2019/12/03–2024/12/26	
	j. Lingjiao Lake Park	469	2019/01/11–2024/12/26	
	k. Huanzi Lake Park	447	2019/01/23–2024/12/26	
	l. Northwest Lake Park	391	2019/01/05–2024/12/27	
	m. Tangxun Lake Park	383	2019/01/27–2024/12/30	
	n. Simeitang Park	373	2019/01/22–2024/12/30	

Note: “Jiangtan” refers to Wuhan's characteristic riverfront embankment spaces that integrate flood control, recreation, and urban landscape functions.

environmental conditions with health outcomes (Liu et al., 2019; Murrin et al., 2023). Therefore, reliance solely on objective environmental indicators may not sufficiently explain why different waterfront spaces generate differentiated well-being experiences, nor reveal how experiential factors are associated with SWB. This highlights the need to move beyond objective environmental assessments and examine how multidimensional experiences, including perceptual, cognitive, and behavioral dimensions, are associated with SWB (Y. Cao et al., 2025; Ogawa et al., 2024).

Finally, existing studies still lack a systematic analytical framework capable of identifying waterfront experiences from large-scale user-generated content and further revealing their associations with SWB. Although the integration of social media data and deep learning techniques has opened new analytical pathways for related research, current studies have mainly focused on applying general NLP models for sentiment analysis (Cheng et al., 2021; Huai et al., 2023; Wei et al., 2024) or environmental satisfaction assessment (Huai et al., 2023; Li et al., 2024; Tang et al., 2023), while the identification of specific public environmental experiences and the processes through which they relate to well-being outcomes remain relatively limited. Some studies have further employed keyword matching and dictionary-based approaches to extract environmental experience-related factors. Nevertheless, such methods rely heavily on predefined rules and may fail to adequately capture complex semantic relationships, contextual information, and multidimensional experiential expressions embedded in natural language (I. Kong et al., 2023; Zheng et al., 2024). Therefore, leveraging advanced deep learning approaches to automatically identify public perceptual, cognitive, and behavioral experiences from large-scale textual data and uncover their associations with SWB remains an important research challenge.

To address the above research gaps, this study takes typical waterfront spaces in Wuhan, China, as the research objects and employs social media review data combined with NLP and machine learning techniques. From a public perspective, the study conducts a fine-grained identification of individuals' perceptual, cognitive, and behavioral characteristics in waterfront environments and further explores how these features are associated with SWB in waterfront environments. By comprehensively analyzing both overall and typology-specific variations in SWB and experiential characteristics, this study provides new theoretical insights and empirical evidence for the optimization of waterfront design and health-oriented urban planning. It also seeks to address three key research questions:

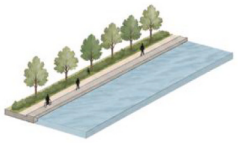
(1) How can deep learning models be utilized to achieve a fine-grained identification of public experiences in waterfront spaces?
(2) How is SWB in waterfront environments associated with multi-dimensional features?
(3) What significant differences exist in the experiential association patterns and seasonal rhythms across different waterfront typologies in relation to SWB?

2. Data collection and preprocessing

2.1. Study area

Wuhan, the capital of Hubei Province and a major national hub city, had an urban population of approximately 13.6 million in 2022 (Yuan et al., 2025). Situated at the confluence of the Yangtze River and its largest tributary, the Han River, and dotted with 166 lakes, the city is known as the “City of a Hundred Lakes” (Yongli et al., 2014). This hydrological configuration, “two rivers converging, lakes encircling”, has forged a prototypical waterfront urban form. Prior studies have documented close links between Wuhan's waterfront spaces, residents' health, and social interaction, providing a solid empirical basis for the present case study (J. Wu et al., 2019; Xie et al., 2022). Given its extensive blue-space resources, distinctive hydromorphological configuration, and strong representativeness in waterfront research, Wuhan serves as an exemplary case city for this study. The Third Ring Road of Wuhan broadly delineates the boundary between the urban core and the suburban areas. Within the Third Ring, the city exhibits higher development intensity and a greater concentration of urban functions and public facilities, whereas areas beyond the ring are relatively less developed (Y. Chen et al., 2022b). Accordingly, this study confines its scope to the waterfront spaces located within the Third Ring Road (Fig. 1).

表 1
14 个滨水研究样地的数据。

类型	网站	评论数 (n)	周期	空间形态
滨水绿道（8643）	a. 东湖绿道	8643	2019/01/14–2024/12/30	 线性
河畔（4936）	b. 汉口江滩	1553	2019/02/06–2024/12/30	
	c. 武昌江滩	1314	2019/01/12–2024/12/24	
	d. 青山江滩	1185	2019/01/26–2024/12/23	
	e. 杭州江滩	884	2019/01/03–2024/12/26	腰带
滨水公园（6841）	f. 沙湖公园	2356	2019/01/05–2024/12/27	
	g. 资阳湖公园	1263	2019/02/08–2024/12/27	
	h. 莫水湖公园	603	2019/01/05–2024/12/29	
	i. 月亮湖公园	556	2019/12/03–2024/12/26	
	j. 灵桥湖公园	469	2019/01/11–2024/12/26	
	k. 环子湖公园	447	2019/01/23–2024/12/26	
	l. 北西湖公园	391	2019/01/05–2024/12/27	
	m. 塘汛湖公园	383	2019/01/27–2024/12/30	
	n. 西梅唐公园	373	2019/01/22–2024/12/30	Areal

注：“江滩”指武汉独具特色的滨河岸线空间，其功能涵盖防洪、休闲及城市景观三大方面。

2023）。因此，仅依赖客观环境指标，可能不足以充分解释为何不同滨水空间会引发差异化的福祉体验，也无法揭示体验性因素与 SWB 之间的关联。这凸显了超越客观环境评估的必要性，转而探究包括感知、认知及行为维度在内的多维体验如何与 SWB 相关联（Cao等，2025；Ogawa等，2024）。

最后，现有研究仍缺乏一个系统化的分析框架，以从大规模用户生成内容中识别滨水体验，并进一步揭示这些体验与 SWB 之间的关联。尽管社交媒体数据与深度学习技术的结合为相关研究开辟了新分析路径，但当前研究主要集中于应用通用NLP模型进行情感分析（Cheng等，2021；Huai等，2023；Wei等，2024）或环境满意度评估（Huai 等，2023；Li 等，2024；Tang 等，2023），而针对特定公共环境体验及其与福祉结果关联机制的识别研究仍相对有限。部分研究还采用了关键词匹配及基于词典的方法来提取与环境体验相关的因素。然而，这些方法高度依赖于预定义的规则，可能无法充分捕捉自然语言中蕴含的复杂语义关系、上下文信息及多维体验表达（I.Kong 等，2023；Zheng 等，2024）。因此，利用先进的深度学习技术，从大规模文本数据中自动识别公众的感知、认知及行为体验，并揭示这些体验与 SWB 之间的关联，仍是一项重要的研究挑战。

为填补上述研究空白，本研究以中国武汉的典型滨水空间为研究对象，结合社交媒体评论数据、自然语言处理及机器学习技术展开分析。从公众视角出发，本研究对个体在滨水环境中的感知、认知及行为特征进行了细粒度识别，并进一步探究这些特征与 SWB 之间的关联。通过全面分析 SWB 及体验特征在整体层面与特定类型层面的差异，本研究为滨水空间设计与健康导向型城市规划的优化提供了新的理论见解与实证依据。同时，本研究亦旨在解答以下三个关键问题：

(1) 如何利用深度学习模型，对滨水空间中的公共体验进行细粒度识别？
(2) 在滨水环境中， SWB 与多维特征有何关联？
(3) 在不同滨水空间类型中，与 SWB 相关的体验关联模式及季节性节律存在哪些显著差异？

2. 数据收集与预处理

2.1. 研究区域

武汉市作为湖北省省会及重要国家枢纽城市，2022年城市人口约为1360万（Yuan等，2025）。该市坐落于长江与其最大支流汉江的交汇处，境内分布着166个湖泊，素有“百湖之城”之称（Yongli等，2014）。这种“双河交汇、湖泊环抱”的水文格局，塑造出典型的滨水城市形态。既往研究已证实武汉滨水空间、居民健康及社会互动之间存在密切关联，为本案例研究提供了坚实的实证基础（J.Wu等，2019；Xie等，2022）。鉴于其丰富的蓝色空间资源、独特的水文地貌特征以及在滨水研究中的高度代表性，武汉成为本研究的典范城市。武汉第三环路大致勾勒出城市核心区与郊区区域的边界。在第三环路区域内，城市呈现出更高的开发强度以及更集中的城市功能与公共设施分布，而环路以外的区域则相对欠发达（Y.Chen et al.，2022b）。因此，本研究将研究范围限定于第三环路内的滨水空间（图 1）。

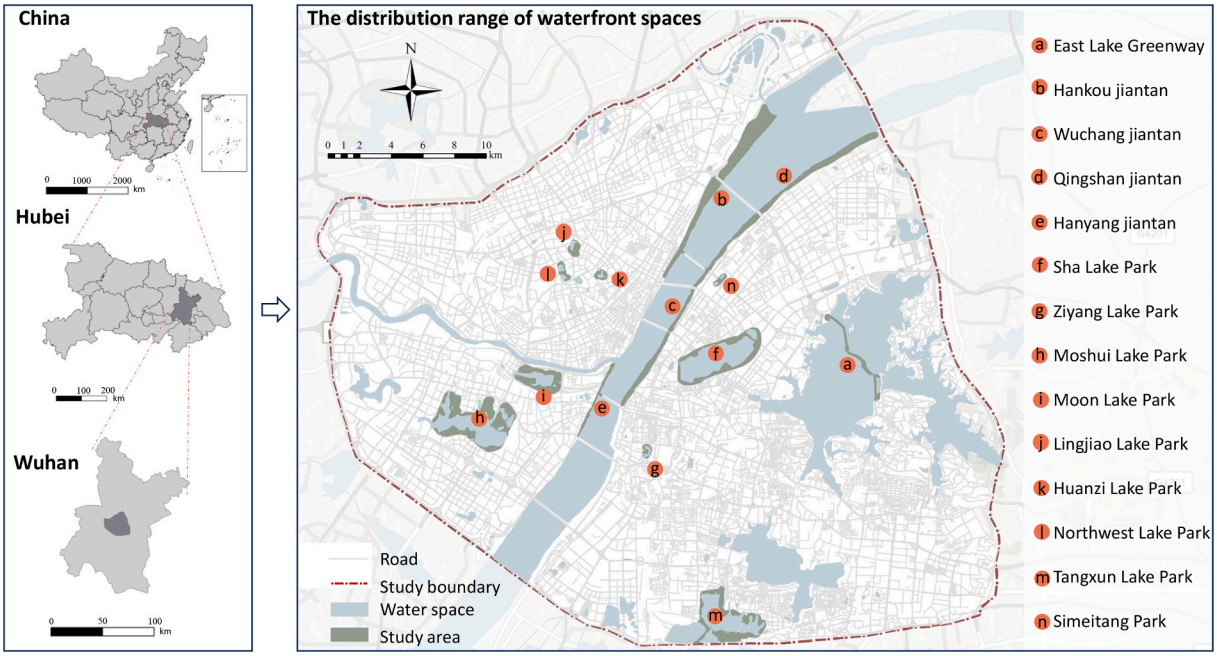


Fig. 1. Study area. Basemap and administrative boundaries: Tianditu Administrative Division Visualization Service (Map approval number: GS (2024) 0650). Road-network data: OpenStreetMap contributors under the Open Database License (ODbL).

2.2. Data sources and processing

As one of China's leading local lifestyle platforms, Dianping aggregates a large volume of user-generated content that captures users' subjective evaluations and experiences of urban spaces, providing an important data foundation for this study (Yuan et al., 2025). This study focuses on urban waterfront spaces in Wuhan, China, and collects user review data from the Dianping platform. Initially, 33,760 reviews from 35 waterfront spaces were obtained, covering the period from 2006 to 2024. Considering user privacy protection requirements, this study only used publicly available and anonymized information relevant to spatial experience analysis, including place labels, publication dates, and review texts. No personally identifiable information was collected or analyzed. To improve data quality and sample representativeness, a multi-stage quality control framework was implemented, including spatial sample selection, temporal filtering, text preprocessing, and noise elimination. First, regarding spatial sample selection, only waterfront spaces with at least 400 raw reviews (prior to data cleaning) were retained to ensure sufficient data availability for subsequent analysis. This threshold helped exclude locations with limited visitor activity and insufficient user feedback, thereby reducing the instability of text-based analyses caused by small sample bias. After this screening process, 14 waterfront spaces were selected as study sites (Fig. 1). These sites cover major waterfront areas along the Yangtze River and Han River, as well as urban lakes of different scales, providing a representative sample of the diverse spatial forms and functional characteristics of Wuhan's waterfront environments. Second, temporal filtering was conducted. Because the number of reviews before 2019 was relatively limited, and data accumulation patterns and platform usage behaviors varied substantially across earlier years, only reviews published between 2019 and 2024 were retained for subsequent analysis. This period provided a more consistent temporal basis for examining seasonal variations and temporal dynamics in waterfront experiences. Finally, systematic text cleaning and preprocessing were conducted. Specifically, duplicate reviews and texts lacking meaningful semantic information were removed, including content consisting solely of emojis, punctuation marks, numerical strings, corrupted characters, and advertisements. Chinese text segmentation was then performed using the Jieba tokenizer. A customized stop-word list was further developed based on word-frequency statistics to filter out auxiliary words, informal online expressions, and symbolic information with limited semantic contribution. Through this multi-stage quality control process, a final dataset of 20,420 valid reviews was obtained, providing a reliable data foundation for subsequent analysis (Table 1; see Appendix A for data preprocessing details).

3. Method

To systematically characterize public experiences in waterfront spaces and further investigate the associations between experiential factors and SWB, this study develops an integrated analytical framework combining deep learning and explainable machine learning (Fig. 2). The framework consists of five major steps. First, based on social media review data, text preprocessing and optimization of semantic embedding models were conducted to construct a high-quality textual corpus and generate semantic representations capturing contextual information embedded in reviews. Second, topic modeling and semantic clustering approaches were employed to identify public waterfront experience themes from large-scale review texts. These themes were further refined through semantic calibration and manual validation to construct a structured experiential indicator system, enabling the transformation of unstructured textual information into multidimensional experiential indicators. Third, an attention-weighted method based on semantic similarity was developed to quantitatively evaluate the semantic associations between review texts and experiential indicators. Fourth, sentiment analysis models were applied to identify emotional expressions in review texts and extract emotion labels and sentiment scores as indicators of experienced well-being. Finally, explainable machine learning models were employed to examine the associations between experiential factors and SWB, identifying key contributing factors and their interaction effects. Through this analytical framework, this study establishes a complete workflow from public experience identification, indicator construction, and emotional quantification to the interpretation of SWB associations, providing systematic empirical evidence for understanding the relationship between urban waterfront environments and public well-being.

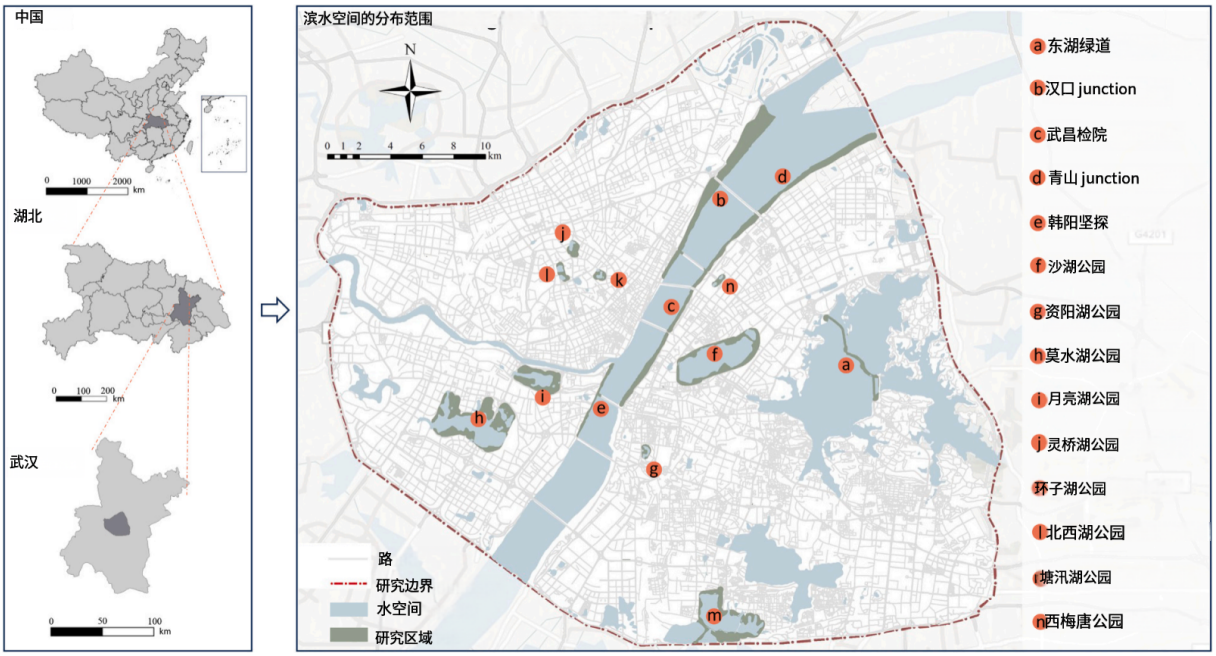


图 1.研究区域。底图及行政区划边界：天地图行政区划可视化服务平台（地图批准文号：GS (2024) 0650）。道路网络数据：OpenStreetMap 贡献者，基于开放数据库许可协议 (ODbL) 提供。

2.2. 数据来源与处理

作为中国领先的本地生活方式平台之一，大众点评汇集了大量用户生成内容，这些内容反映了用户对城市空间的主观评价与体验，为本研究提供了重要的数据基础（Yuan 等，2025）。本研究聚焦中国武汉的城市滨水空间，从大众点评平台收集用户评论数据。最初，本研究从35个滨水空间获取了33,760条评论，时间跨度为2006年至2024年。考虑到用户隐私保护要求，本研究仅使用与空间体验分析相关的公开可用且已匿名化的信息，包括地点标签、发布日期及评论文本。未收集或分析任何个人可识别信息。为提高数据质量与样本代表性，本研究实施了多阶段质量控制框架，包括空间样本选择、时间过滤、文本预处理及噪声消除。首先，关于空间样本筛选，为确保后续分析具备充足的数据基础，本研究仅保留拥有至少400条原始评论（数据清洗前）的滨水空间。该筛选标准有助于排除游客活动量有限且用户反馈不足的场所，从而降低因样本量过小导致的文本分析结果不稳定性问题。经此筛选流程后，最终选定14处滨水空间作为研究样本点（图1）。这些样本点涵盖长江与汉江沿岸的主要滨水区域，以及不同规模的城市湖泊，为武汉市滨水环境的多样化空间形态与功能特征提供了代表性样本。其次，本研究实施了时间筛选：鉴于2019年前发布的评论数量相对较少，且早期年份的数据积累模式与平台使用行为存在显著差异，后续分析仅保留2019年至2024年间发布的评论。该时间段为探究滨水体验的季节性变化与时间动态特征提供了更为一致的时序基础。最后，我们进行了系统的文本清洗与预处理。具体而言，剔除了重复评论及缺乏有效语义信息的文本，包括仅由表情符号、标点符号、数字字符串、乱码及广告构成的内容。随后，采用Jieba分词器对中文文本进行分词处理。基于词频统计结果，进一步构建了定制化停用词列表，以过滤掉助词、非正式网络用语以及语义贡献有限的符号信息。通过这一多阶段质量控制流程，最终获得包含20,420条有效评论的数据集，为后续分析提供了可靠的数据基础（表1；数据预处理细节参见附录A）。

3. 方法

为了系统性地刻画公众在水滨空间中的体验，并进一步探究体验因素与 SWB 之间的关联，本研究开发了一个整合深度学习与可解释机器学习的方法框架（图2）。该框架包含五个主要步骤：首先，基于社交媒体评论数据，进行文本预处理及语义嵌入模型的优化，以构建高质量文本语料库，并生成能够捕捉评论中蕴含的上下文信息的语义表征；其次，采用主题建模与语义聚类方法，从大规模评论文本中识别公众在水滨空间的体验主题。这些主题随后通过语义校准与人工验证进一步精炼，从而构建出结构化的体验指标体系，将非结构化文本信息转化为多维体验指标；最后，开发了一种基于语义相似性的注意力加权方法，用于定量评估评论文本与体验指标之间的语义关联。第四，本研究应用情感分析模型来识别评论文本中的情感表达，并提取情感标签与情感评分，作为个体福祉体验的指标。最后，采用可解释机器学习模型来探究体验性因素与 SWB 之间的关联，识别关键影响因素及其交互效应。通过这一分析框架，本研究构建了从公共体验识别、指标构建、情感量化到 SWB 关联解读的完整工作流程，为理解城市滨水环境与公众福祉之间的关系提供了系统的实证依据。

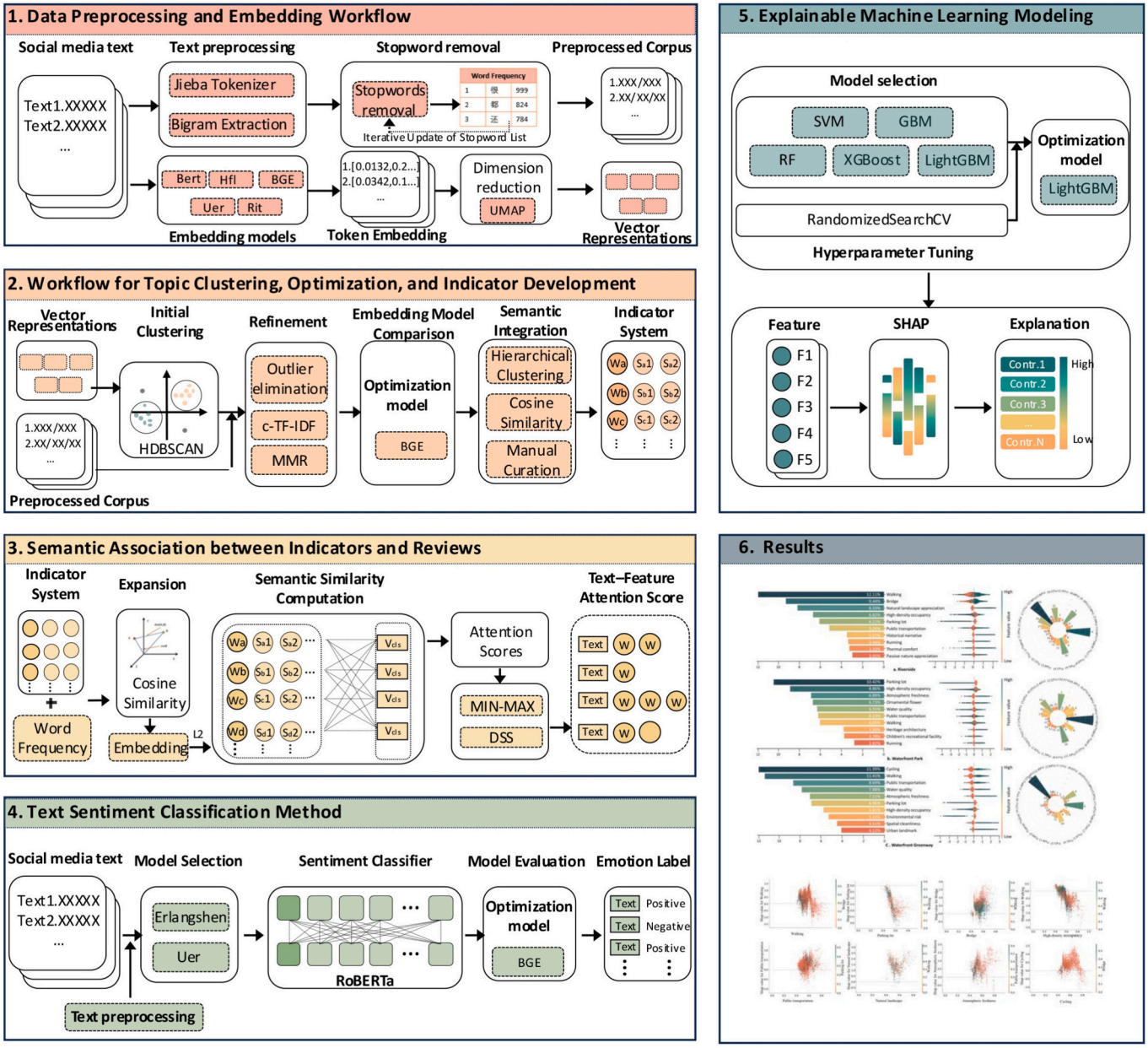


Fig. 2. Research framework.

3.1. Data preprocessing and embedding process

BERTopic is a topic modeling framework built upon Bidirectional Encoder Representations from Transformers (BERT), integrating deep semantic embeddings with topic extraction techniques to adaptively identify distinct semantic categories and extract representative keywords (Raman et al., 2024). To ensure high-quality topic discovery and semantic differentiation, this study performed targeted module optimization and parameter tuning within the BERTopic framework, tailored to the characteristics of waterfront space research and the specific analytical requirements of this study.

We systematically preprocessed the raw review texts to ensure the quality and consistency of the corpus. The jieba tool was used for Chinese word segmentation, and a bigram model was introduced to link high-frequency word pairs, enhancing semantic coherence. Subsequently, considering the significant impact of segmentation quality on topic modeling and clustering results, we optimized the stopword list through word frequency analysis to remove irrelevant or redundant terms. After several rounds of filtering, the resulting corpus preserved the core semantics of waterfront spaces while effectively reducing noise, providing a reliable input for subsequent modeling.

The quality of text embeddings directly influences the effectiveness of topic clustering. To comprehensively capture the deep semantic features contained in the review texts, five embedding models were selected, compared, and evaluated based on their semantic expressiveness, adaptability to Chinese corpora, and embedding stability. Subsequently, Uniform Manifold Approximation and Projection (UMAP) was applied for dimensionality reduction, aiming to preserve the global semantic structure while improving clustering and visualization performance. The resulting low-dimensional semantic space not only facilitates the intuitive visualization of semantic distributions but also establishes a solid foundation for subsequent topic modeling and indicator system construction (see Appendix B for details).

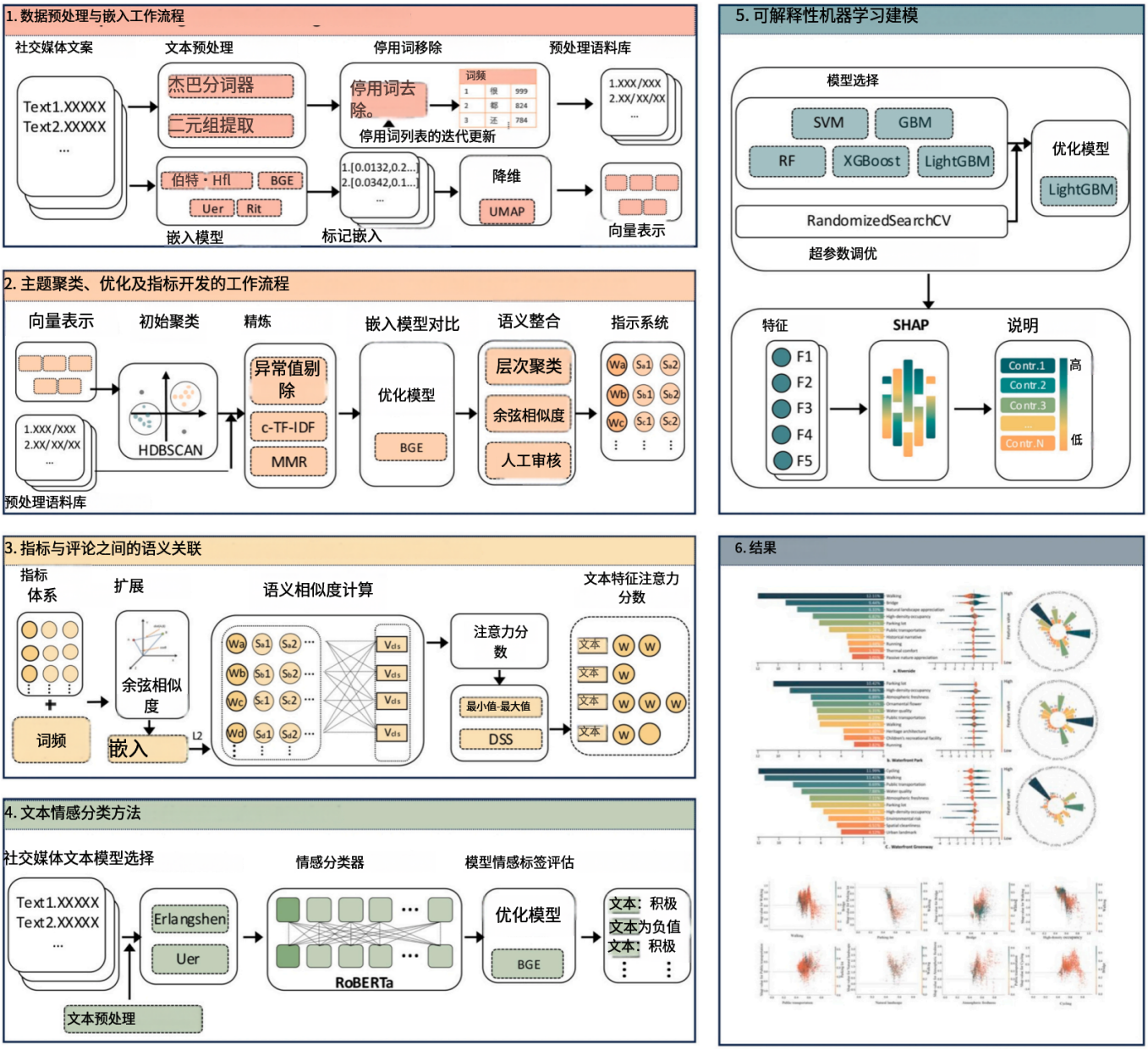


图 2.研究框架。

3.1. 数据预处理与嵌入流程

BERTopic 是一个基于 Transformer 双向编码器表示（BERT）构建的主题建模框架，它将深度语义嵌入与主题提取技术相结合，以自适应方式识别不同的语义类别并提取代表性关键词（Raman 等，2024）。为确保高质量的主题发现与语义区分，本研究在 BERTopic 框架内进行了针对性的模块优化与参数调优，这些优化与调优均针对滨水空间研究的特性及本研究的特定分析需求而设计。

为确保语料库的质量与一致性，我们对原始评论文本进行了系统性预处理。首先采用 jieba 工具进行中文词法分割，并引入二元组模型以关联高频词对，从而增强语义连贯性。鉴于词法分割质量对主题建模及聚类结果具有重要影响，我们基于词频分析优化了停用词列表，剔除无关或冗余词汇。经过多轮筛选后，最终语料库在有效减少噪声的同时，完整保留了滨水空间的核心语义特征，为后续建模提供了可靠的基础数据。

文本嵌入的质量直接影响主题聚类的效果。为了全面捕捉评论文本中蕴含的深层语义特征，本研究选取了五种嵌入模型，并根据其语义表达能力、对中文语料库的适应性以及嵌入稳定性进行了比较与评估。随后，采用均匀流形逼近与投影（UMAP）方法进行降维处理，旨在保留全局语义结构的同时，提升聚类与可视化性能。所得的低维语义空间不仅有助于直观地可视化语义分布，还为后续的主题建模及指标体系构建奠定了坚实基础（详见附录B）。

3.2. Topic clustering and indicator system construction

The HDBSCAN algorithm was applied for density-based adaptive topic clustering, and the class-based TF-IDF (c-TF-IDF) method was subsequently used to enhance the representativeness and discriminative power of topic keywords. In the clustering process, some texts exhibited weak semantic coherence and could not be assigned to any topic; these were treated as outliers. To improve the stability and interpretability of the clustering results, a cosine similarity–based outlier filtering mechanism with adaptive thresholding was implemented, which effectively reduced noise interference and clarified topic boundaries. Additionally, the Maximal Marginal Relevance (MMR) algorithm was applied to rank topic keywords, maintaining a balance between representativeness and diversity while preventing dominance by high-frequency terms. Parameter tuning and performance comparisons were conducted across different embedding models using UMAP visualization, through which BAAI/bge-large-zh-v1.5 was identified as the optimal embedding model for semantic clustering. This model subsequently served as the foundation for semantic calibration and indicator system construction (see Appendix C for technical details and clustering results).

Although unsupervised clustering can extract semantic categories from the public perspective, its interpretability and accuracy remain limited. Therefore, this study proposed a multi-level semantic calibration strategy. First, cosine similarity between word embeddings was applied to merge semantically similar keywords, thereby improving topic coherence and compactness (Eq. 1). Second, the clustering results were manually reviewed and semantically refined. To minimize subjective bias, a multi-coder cross-validation procedure was adopted. Researchers (KW and DX) independently organized and integrated the identified topics, while researchers (XZ and HY) conducted an independent review and refined topics with ambiguous boundaries or semantic overlap, thereby improving the clarity and interpretability of the final topic classification. Finally, the semantic similarity between topic keywords and their corresponding seed words was calculated, and the average value was defined as the semantic aggregation index, providing a quantitative measure of topic consistency (Eq. 2).

$$\mathrm{CosSim}(w_i, s) = \frac{\vec{v}_i \cdot \vec{v}_s}{\left\|\vec{v}_i\right\| \cdot \left\|\vec{v}_s\right\|} \tag{1}$$

$$Cohesion(T) = \frac{1}{n}\sum_{i=1}^{n} \mathrm{CosSim}(w_i, s) \tag{2}$$

where w_i and s denote a keyword and its corresponding seed word. $\vec{v}_i$ and $\vec{v}_s$ represent their corresponding word embedding. T represents a given topic, and n indicates the number of seed words contained in that topic.

Through the integrated application of semantic fusion, keyword filtering, and semantic aggregation analysis, this study constructed a comprehensive evaluation indicator system from the public perspective. The framework covers three overarching dimensions: Environmental Perception, Environmental Cognition, and Behavioral Activity, comprising 21 subdimensions and 63 indicators in total (Table 2). Methodologically, the system fully leverages the fine-grained feature extraction capability of BERT-based models, enabling a more comprehensive and accurate representation of the public's multifaceted experiences in waterfront spaces. Moreover, it provides a solid methodological foundation for subsequent analyses (see Appendix D for detailed definitions and descriptions of each indicator).

Table 2
Structured indicator framework.

Dimension	Subdimension	Indicator keywords
a. Environmental perception	1. Vegetation Landscape Perception	Tree (0.76), Grassland (0.71), Ornamental flower (0.75)
	2. Animal Diversity Perception	Bird (0.76), Aquatic animal (0.70), Insect (0.76), Pet (0.70)
	3. Architectural Landscape Perception	Pavilion (0.75), High-rise building (0.80), Bridge (0.74), Corridor (0.72), Platform (0.73), Stair (0.78)
	4. Environmental Cleanliness Perception	Spatial cleanliness (0.81), Water quality (0.72)
	5. Public Service Facilities Perception	Seating facility (0.74), Sanitation facility (0.80), Recreational facility (0.80), Sports facility (0.78)
	6. Traffic Perception	Parking lot (0.80), Vehicle (0.78), Ferry (0.74), Public transportation (0.75)
	7. Auditory Perception	Natural sound (0.65), Anthropogenic sound (0.67)
	8. Olfactory Perception	Atmospheric freshness (0.76), Odor (0.80)
	9. Thermal Perception	Thermal comfort (0.81), Heat (0.80), Cold (0.76)
	10. Crowding Perception	High crowd density (0.79), Low crowd density (0.76)
b. Environmental cognition	11. Safety Cognition	Flood risk (0.70), Environmental risk (0.75), Night illumination (0.69)
	12. Supporting Services Cognition	Catering service (0.72), Commercial service (0.75), Accommodation service (0.74)
	13. Spatial Structure Cognition	Spatial morphology (0.80), Spatial configuration (0.76)
	14. Spatial Design Cognition	Waterside-oriented design (0.74), Creative spatial design (0.72), Stylistic characteristic (0.75)
	15. Spatial Renewal Cognition	Construction and renovation (0.80)
	16. Spatial Image Cognition	Urban landmark (0.77), Site-specific attraction (0.74)
	17. Cultural and Emotional Cognition	Historical narrative (0.70), Heritage architecture (0.84), Emotional memory (0.72)
c. Behavioral activity	18. Leisure and Relaxation Activity	Passive nature appreciation (0.78), Camping and outdoor dining (0.75), Recreational engagement (0.83), Photographic engagement (0.77)
	19. Physical Activity	Dance (0.75), Cycling (0.79), Running (0.78), Ball game (0.76), Walking (0.79)
	20. Social Interaction Activity	Parent–child interaction (0.76), Couple interaction (0.71), Group cohesion activity (0.78)
	21. Observational Experience Activity	Natural landscape appreciation (0.71), Cultural landscape appreciation (0.79)

3.3. Evaluation of semantic associations between the indicator system and review texts

To quantitatively assess the semantic relationships between the constructed indicator system and reviews, this study proposes an attention-weighted method based on semantic similarity. The approach integrates semantic expansion, vector representation, and dynamic filtering to achieve high-precision alignment between indicators and text semantics. The procedure consists of four steps.

(1) Lexicon Expansion. For each indicator's seed-word set, semantically similar words were identified from a high-frequency vocabulary based on cosine similarity (Eq. 1), thereby expanding lexical coverage and enhancing the indicator's ability to capture relevant textual content. This process resulted in a comprehensive seed-word set for each indicator.
(2) Semantic Vector Representation. Review texts and keywords were encoded using the BAAI/bge-large-zh-v1.5 embedding model. Mean pooling and L2 normalization were subsequently

3.2. 主题聚类与指标体系构建

采用 hdbscan 算法进行基于密度的自适应主题聚类，随后使用基于类别的 TF-IDF（c-TF-IDF）方法增强主题关键词的代表性和区分能力。在聚类过程中，部分文本表现出较弱的语义连贯性，无法被归入任何主题；这些文本被视为异常值。为了提高聚类结果的稳定性和可解释性，实施了基于余弦相似度的异常值过滤机制，并采用自适应阈值，有效减少了噪声干扰，明确了主题边界。此外，应用最大边际相关性（MMR）算法对主题关键词进行排序，在保持代表性与多样性的平衡的同时，防止高频词占据主导地位。通过 UMAP 可视化，对不同嵌入模型进行了参数调优和性能比较，最终确定 BAAI /bge-large-zh-v1.5 为语义聚类的最佳嵌入模型。该模型随后成为语义校准及指标体系构建的基础（技术细节与聚类结果详见附录 C）。

尽管无监督聚类能够从公共视角提取语义类别，但其可解释性与准确性仍有限。为此，本研究提出了一种多层次语义校准策略。首先，应用词嵌入间的余弦相似度来合并语义相近的关键词，从而提升主题连贯性与紧凑性（等式1）。其次，对聚类结果进行人工审核并优化语义。为减少主观偏差，本研究采用多标注者交叉验证流程：研究人员（KW与DX）独立整理并整合已识别的主题，而研究人员（XZ 与HY）则独立复核并对边界模糊或语义重叠的主题进行优化，进而提升最终主题分类的清晰度与可解释性。最后，计算主题关键词与其对应种子词间的语义相似度，并将该平均值定义为语义聚合指数，从而为主题一致性提供量化度量（等式2）。

$$\mathrm{CosSim}(w_i, s) = \frac{\vec{v}_i \cdot \vec{v}_s}{\left\|\vec{v}_i\right\| \cdot \left\|\vec{v}_s\right\|} \tag{1}$$

$$Cohesion(T) = \frac{1}{n}\sum_{i=1}^{n} \mathrm{CosSim}(w_i, s) \tag{2}$$

其中w_i与s分别表示一个关键词及其对应的种子词；$\rightarrow v_i$与$\rightarrow v_s$分别表示它们对应的词向量；T 表示给定主题，n 表示该主题中包含的种子词数量。

本研究通过整合语义融合、关键词过滤及语义聚合分析等技术，从公众视角构建了一套综合性评价指标体系。该框架涵盖三大核心维度——环境感知、环境认知与行为活动，共包含21个子维度及63项指标（表2）。在方法论层面，本体系充分运用基于BERT模型的细粒度特征提取能力，从而更全面、精准地刻画公众在水滨空间中的多维体验。此外，该体系还为后续分析提供了坚实的方法学基础（各指标的详细定义与说明参见附录D）。

3.3. 指标体系与综述文本间语义关联性的评估

为定量评估所构建的指标体系与评论之间的语义关联，本研究提出了一种基于语义相似性的注意力加权方法。该方法整合语义扩展、向量表示与动态过滤，从而实现指标与文本语义之间的高精度对齐。该流程包含以下四个步骤。

(1) 词典扩展。针对每个指标的种子词集，我们从高频词汇库中基于余弦相似度（等式1）识别出语义相近的词汇，从而扩展词汇覆盖范围，并增强该指标捕捉相关文本内容的能力。此过程为每个指标最终构建了一个全面的种子词集。

(2) 语义向量表示。使用 BAAI /bge-large-zh-v1.5 嵌入模型对文本及关键词进行编码。随后应用均值池化与 L2 归一化处理，将所有嵌入

表 2
结构化指标框架。

维度	子维度	指示性关键词
a. 环境感知	1. 植被景观感知	乔木（0.76）、草地（0.71）、观赏花卉（0.75）
	2. 对动物多样性的认知	鸟类 (0.76)、水生动物 (0.70)、昆虫 (0.76)、宠物 (0.70)
	3. 建筑景观感知	凉亭（0.75）、高层建筑（0.80）、桥梁（0.74）、走廊（0.72）、平台（0.73）、楼梯（0.78）、空间整洁度（0.81）、水质（0.72）
	4. 环境清洁感知	
	5. 公共服务设施认知	就座设施（0.74）、卫生设施（0.80）、休闲设施（0.80）、体育设施（0.78）
	6. 交通感知	停车场 (0.80)、车辆 (0.78)、渡轮 (0.74)、公共交通 (0.75)
	7. 听觉感知	自然声（0.65）、人为声（0.67）、空气清新度（0.76）、气味（0.80）
	8. 嗅觉感知	
	9. 温度感知	热舒适度（0.81）、炎热感（0.80）、寒冷感（0.76）
	10. 拥挤感知	人群密度高（0.79），人群密度低（0.76）
	11. 安全认知	洪水风险（0.70）、环境风险（0.75）、夜间照明（0.69）
b. 环境认知	12. 辅助服务认知	餐饮服务（0.72）、商业服务（0.75）、住宿服务 服务（0.74）
	13. 空间结构认知	空间形态（0.80）、空间构型（0.76）
	14. 空间设计认知	滨水导向设计（0.74）、创意空间设计（0.72） 风格特征（0.75）
	15. 空间更新认知	建筑及装修工程（0.80）
	16. 空间图像认知	城市地标性景观（0.77）、场地特定吸引力（0.74）
	17. 文化与情感认知	历史叙事（0.70）、传统建筑（0.84）、情感记忆（0.72）
c. 行为学活动	18. 休闲放松活动	被动式自然欣赏（0.78）、露营与户外用餐（0.75）、休闲活动参与（0.83）、摄影活动参与（0.77）、舞蹈（0.75）、骑行（0.79）、跑步（0.78）、球类运动（0.76）、步行（0.79）
	19. 体力活动	
	20. 社交互动活动	亲子互动（0.76）、夫妻互动（0.71）、小组凝聚力活动（0.78）
	21. 观察实践活动	自然景观欣赏度（0.71）、文化景观欣赏度（0.79）

applied to project all embeddings into a unified semantic space, ensuring consistency in vector similarity calculations.

(3) Attention Score Calculation. For each indicator, we calculated the cosine similarity between all seed-word embeddings $W_i = \{s_{i1}, s_{i2}, \ldots, s_{im}\}$ and the mean-pooled embedding of the review text v_{CLS}. Specifically, each seed word within an indicator was compared with the review embedding to generate a set of similarity scores reflecting the semantic relevance between the review and the indicator-related concepts. The mean of the three highest similarity values was defined as the attention score $AttentionScore_i$ (Eq. 3), capturing strong semantic correlations while avoiding dilution from low-similarity words. This process produced a set of semantic relevance scores between each review and all indicators, denoted as AttentionScores (Eq. 4). Consequently, each review can be transformed into a multidimensional indicator vector consisting of the AttentionScores of all indicators, where each AttentionScore quantifies the semantic association strength between the review and a given indicator and is treated as a continuous variable in downstream analyses.

(4) Dynamic Selection Strategy. To avoid overmatching in short texts and information dilution in long texts, this study introduced a dynamic filtering strategy based on min–max normalization (Eq. 5), which adaptively controlled the number of retained indicators according to text length. This strategy ensured that each review was represented only by the most representative indicators, thereby improving the precision and robustness of semantic matching (see Appendix E for details).

$$AttentionScore_i = \frac{1}{k}\sum_{j\in \mathrm{Top}-k}\left(e_{ji}\cdot v_{\mathrm{CLS}}\right) \tag{3}$$

$$\mathrm{AttentionScores} = \{\mathrm{AttentionScore}_1, \ldots, \mathrm{AttentionScore}_n\} \tag{4}$$

$$x_{\mathrm{new}} = \frac{x - \min(x)}{\max(x) - \min(x)} \tag{5}$$

3.4. Measurement of subjective well-being

In this study, SWB is operationalized based on sentiment-derived emotional expressions from social media data, capturing its affective dimension. RoBERTa is an improved variant of the BERT pre-trained language model that enhances training performance through a larger corpus, extended training duration, and optimized training strategies. It has demonstrated superior accuracy and generalization capabilities across various NLP tasks, particularly in sentiment classification (Briskilal and Subalalitha, 2022). To ensure accuracy and applicability in Chinese sentiment analysis, this study compared two open-source RoBERTa-based models pre-trained on large-scale Chinese corpora: IDEA-CCNL/Erlangshen-RoBERTa-110 M-Sentiment (Erlangshen) and UER/roberta-base-finetuned-dianping-chinese (UER). A total of 1500 review samples were randomly selected for manual annotation to evaluate the models' classification performance on positive and negative emotions. Five standard evaluation metrics were employed: Accuracy, Precision, Recall, F1-score, and AUC (Area Under the ROC Curve). The evaluation results are shown in Table 3. Although the UER model exhibits a slightly lower Recall compared with the Erlangshen model, it outperforms Erlangshen on the key evaluation metrics of Accuracy, Precision, F1-score, and AUC, demonstrating superior overall performance and greater stability. Therefore, the UER model was adopted as the primary sentiment classification tool in this study. Based on its output, standardized emotion labels and emotion scores were generated for subsequent quantitative analyses. The emotion labels were treated as binary categorical variables (see Appendix F for details).

In addition, following previous studies (Gao and Fang, 2025; Tu et al., 2026), we introduce the Happiness Intensity Index (HII) as an operational proxy for the affective dimension of SWB (Eq. 6). The index measures the overall intensity of happiness for a given space by calculating the proportion of positive reviews to total reviews within a specified time interval *T*, thereby reflecting the general happiness level in waterfront environments. HII ranges from 0 to 1 and represents a continuous measure of happiness intensity. In this context, "happiness" refers specifically to the affective component of SWB captured by HII. Where N_{positive} and N denote the numbers of positive reviews and total reviews, respectively.

$$HII_{At} = \frac{N_{positive}}{N} \tag{6}$$

3.5. Explainable machine learning modeling

After completing text feature extraction and sentiment classification, the dataset was partitioned into training, validation, and test sets in a 64:16:20 ratio using stratified sampling. To ensure representative and stable model selection, five typical machine learning algorithms were compared in terms of performance: Support Vector Machine (SVM), Gradient Boosting (GB), Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Light Gradient Boosting Machine (LightGBM). RandomizedSearchCV was employed to perform randomized hyperparameter search within predefined parameter spaces, evaluating 15–25 hyperparameter combinations for each model. A three-fold stratified cross-validation procedure was used to further optimize model performance and generalization ability. To mitigate the effect of class imbalance on performance evaluation, this study adopted macro-averaging to compute five metrics: Accuracy, Precision, Recall, F1-score, and AUC. As shown in Table 4, LightGBM achieved the best overall performance across multiple metrics, demonstrating both strong discriminative power and high stability. Therefore, LightGBM was selected as the core model for subsequent feature importance analysis (see Appendix G for details).

To further interpret the model's internal mechanisms and enhance result transparency (T. Hu et al., 2023), this study introduces the Shapley Additive exPlanations (SHAP) method, which improves the model's interpretability from a game-theoretic perspective. In this study, each input feature is considered a participant in the game, with the goal of collectively determining the prediction output. Shapley values (Eq. 7) are used to measure each feature's fair contribution to the prediction, reflecting the marginal contribution of the feature to the prediction for an individual sample. Additionally, Shapley interaction values (Eq. 8) further decompose the joint contribution of feature pairs, revealing the interaction relationships between different features (I. Kong et al., 2023).

$$\Phi_i = \sum_{S\subseteq N\setminus\{i\}} \frac{|S|!(M-|S|-1)!}{M!}\left[f_x(S\cup\{i\}) - f_x(S)\right] \tag{7}$$

Table 3
Sentiment model evaluation results.

Model	Accuracy	Precision	Recall	F1-score	AUC
Erlangshen	0.9067	0.8867	0.9236	0.9048	0.9073
UER	0.9333	0.9379	0.9222	0.9300	0.9329

Table 4
Machine learning model evaluation results.

Model	Accuracy	Precision	Recall	F1-score	AUC
SVM	0.87	0.68	0.63	0.65	0.71
Gradient Boosting	0.88	0.71	0.64	0.67	0.78
Random Forest	0.87	0.69	0.68	0.69	0.78
XGBoost	0.87	0.70	0.68	0.69	0.77
LightGBM	0.88	0.71	0.68	0.69	0.78

向量投影至统一的语义空间，以确保向量相似度计算的一致性。

(3) 注意力分数计算。对于每个指标，我们计算了所有种子词嵌入向量$W_i = \{s_{i1}, s_{i2}, \ldots, s_{im}\}$与评论文本的平均池化嵌入向量$v_{CLS}$之间的余弦相似度。具体而言，将每个指标中的种子词与评论嵌入向量进行比较，从而生成一组相似度分数，这些分数反映了评论与指标相关概念之间的语义关联度。将三个最高相似度值的平均值定义为注意力分数$AttentionScore_i$（等式3），该方法既能捕捉强语义关联，又可避免因低相似度词汇而导致的分数稀释效应。此过程为每篇评论与所有指标生成了一个语义关联分数集合，记为AttentionScores（等式4）。由此，每篇评论可被转换为一个由所有指标的AttentionScores构成的多维指标向量，其中每个AttentionScore量化了该评论与特定指标之间的语义关联强度，并在后续分析中作为连续变量处理。

(4) 动态选择策略。为避免短文中出现过度匹配以及长文中出现信息稀释现象，本研究引入了一种基于极值归一化（等式5）的动态过滤策略，该策略可根据文本长度自适应地控制保留指标的数量。此策略确保每篇评论仅由最具代表性的指标表征，从而提升语义匹配的精确度与鲁棒性（详见附录E）。

$$AttentionScore_i = \frac{1}{k}\sum_{j\in \mathrm{Top}-k}\left(e_{ji}\cdot v_{\mathrm{CLS}}\right) \tag{3}$$

$$\mathrm{AttentionScores} = \{\mathrm{AttentionScore}_1, \ldots, \mathrm{AttentionScore}_n\} \tag{4}$$

$$x_{\mathrm{new}} = \frac{x - \min(x)}{\max(x) - \min(x)} \tag{5}$$

3.4. 主观幸福感的测量

在本研究中，SWB是基于社交媒体数据中的情感表达而定义的，旨在捕捉其情感维度。RoBERTa是BERT预训练语言模型的改进版本，通过更大的语料库、更长的训练时间和优化的训练策略来提升训练性能。它在多种自然语言处理任务中均展现出卓越的准确性和泛化能力，尤其是在情感分类任务中（Briskilal 和 Subalalitha，2022）。为确保在中国情感分析中的准确性和适用性，本研究比较了两种基于RoBERTa的开源模型，这些模型均使用大规模中文语料库预训练：IDEA- CCNL /Erlangshen-RoBERTa-110 M-Sentiment（Erlangshen）和UER /roberta-base-finetuned-dianping-chinese（UER）。共随机选取1500条评论样本进行人工标注，以评估模型对正面和负面情感的分类性能。采用了五项标准评估指标：准确率、精确率、召回率、F1分数以及AUC（ROC曲线下面积）。评估结果如表3所示。尽管UER模型相较于Erlangshen模型在召回率上略低，但在准确率、精确率、F1分数及AUC等关键评估指标上均优于Erlangshen模型，展现出更优的整体性能与更高的稳定性。因此，本研究将UER模型选作主要的情感分类工具。基于其输出结果，我们生成了标准化的情感标签与情感分数，用于后续定量分析。这些情感标签被视作二分类变量（详见附录F）。

表3
情感模型评估结果。

模型	精度	准确率	召回率	F1分数	AUC
Erlangshen	0.9067	0.8867	0.9236	0.9048	0.9073
UER	0.9333	0.9379	0.9222	0.9300	0.9329

此外，基于既往研究（Gao 与 Fang，2025；Tu等，2026），我们引入“幸福强度指数”（HII）作为SWB情感维度的操作性代理指标（等式6）。该指数通过计算特定时间区间T内正面评论数占评论总数的比例，衡量给定空间内幸福感的整体强度，从而反映滨水环境中的总体幸福水平。HII取值范围为0至1，代表幸福强度的连续度量。在此语境下，“幸福感”特指HII所捕捉的SWB的情感成分。其中N_{positive}与N分别表示正面评论数与评论总数。

$$HII_{At} = \frac{N_{positive}}{N} \tag{6}$$

3.5. 可解释机器学习建模

在完成文本特征提取和情感分类后，数据集采用分层抽样法按64:16:20的比例划分为训练集、验证集和测试集。为确保模型选择的代表性与稳定性，本研究比较了五种典型机器学习算法的性能：支持向量机（SVM）、梯度提升（GB）、随机森林（RF）、极端梯度提升（XGBoost）以及轻量梯度提升机（LightGBM）。使用`RandomizedSearchCV`在预定义的参数空间中进行随机超参数搜索，为每个模型评估15–25种超参数组合。此外，采用三折分层交叉验证流程以进一步优化模型性能及泛化能力。为减轻类别不平衡对性能评估的影响，本研究采用宏平均法计算五项指标：准确率、精确率、召回率、F1分数及AUC值。如表4所示，LightGBM在多项指标上均取得最佳整体性能，既展现出强大的判别能力，又具备高稳定性。因此，LightGBM被选作后续特征重要性分析的核心模型（详见附录G）。

为进一步阐释模型的内部机制并提升结果的可解释性（T.Hu et al.，2023），本研究引入了Shapley Additive exPlanations（Shap）方法，该方法从博弈论视角提升了模型的可解释性。在本研究中，每个输入特征均被视为该博弈中的参与者，其共同目标是确定预测输出结果。采用Shapley值（等式7）来量化每个特征对预测结果的公平贡献，反映该特征对单个样本预测结果的边际贡献。此外，Shapley交互值（等式8）可进一步分解特征对的联合贡献，揭示不同特征间的交互关系（I.Kong et al.，2023）。

$$\Phi_i = \sum_{S\subseteq N\setminus\{i\}} \frac{|S|!(M-|S|-1)!}{M!}\left[f_x(S\cup\{i\}) - f_x(S)\right] \tag{7}$$

表4
机器学习模型评估结果。

模型	精度	准确率	召回率	F1分数	AUC
SVM	0.87	0.68	0.63	0.65	0.71
梯度提升	0.88	0.71	0.64	0.67	0.78
随机森林	0.87	0.69	0.68	0.69	0.78
XGBoost	0.87	0.70	0.68	0.69	0.77
LightGBM	0.88	0.71	0.68	0.69	0.78

$$\Phi_{ij}(f,x) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(M-|S|-2)!}{2(M-1)!} \nabla_{ij}(S) \quad (8)$$

where φ_i and φ_{ij} denote the Shapley value and Shapley interaction value, respectively; M is the total number of features, N is the feature set, and S is a subset of N excluding feature i.

4. Results

4.1. Characterizing HII patterns of urban waterfronts

4.1.1. Overall HII patterns

When all data across different waterfront types were aggregated, the overall HII reached 0.882, indicating that the public generally experiences a high level of SWB in waterfront spaces. Weekly aggregation provided finer insights into temporal dynamics. As shown in Fig. 3a, the median HII was 0.90, with an interquartile range (IQR) of 0.85–0.95 and a minimum value of 0.70, suggesting that HII was consistently concentrated within a high range, with only occasional declines. In terms of temporal trends (Fig. 3b), the HII increased steadily during 2019–2020, maintaining a level of around 0.95, then declined markedly between 2020 and 2022, reaching its lowest point (0.80) in mid-2021. Thereafter, it gradually recovered and stabilized above 0.90 from 2023 onward. Short-term fluctuations exhibited a clear seasonal pattern: HII tended to rise slightly or reach short-term peaks in spring (March–May). It then declined markedly in summer (June–August) to a seasonal trough, rebounded in autumn (September–November), and remained relatively high in winter (December–February) despite a slight decrease.

4.1.2. HII patterns across waterfront typologies

Fig. 4 illustrates the overall distribution and temporal variation of HII across different waterfront typologies. Riverfront exhibited the highest median HII (0.92) with a relatively concentrated distribution (0.89–0.95) and a minimum value of 0.80, indicating that public happiness in these spaces generally remained high, with only minor fluctuations and occasional declines. Waterfront parks showed a slightly lower median (0.91) but the most concentrated distribution (0.89–0.94) and a higher minimum value (0.86), suggesting a marginally lower overall level than riverfront areas but greater temporal stability. In contrast, waterfront greenways had the lowest median (0.90) and considerably lower first quartile (0.82) and minimum values (0.70), reflecting weaker and more volatile happiness experiences overall (Fig. 4a). From a temporal perspective, both riverfront and park spaces exhibited relatively stable long-term trends, maintaining consistent HII levels around 0.90–0.93 with limited short-term fluctuations and a generally alternating upward pattern. In comparison, greenways demonstrate the most pronounced volatility, characterized by recurring phases of decline followed by recovery (Fig. 4b). In terms of seasonal patterns, riverfronts displayed distinct seasonal fluctuations: HII peaked in late spring to early summer (May), dropped to a trough (0.87) in late summer (July–August), gradually rebounded during autumn, and declined slightly again in winter (December–January). Parks remained the most stable, maintaining values between 0.90 and 0.93 across most months, with minimal seasonal variation. Greenways, however, exhibited a typical "U-shaped" seasonal pattern—relatively high at the beginning of the year (January–February, around 0.90), declining steadily to a summer low (August, occasionally as low as 0.75–0.80), and then gradually recovering to around 0.90 in autumn and winter (Fig. 4c).

4.2. Relative feature importance analysis based on SHAP

4.2.1. Overall feature importance

Fig. 5 illustrates the feature importance results across multiple subdimensions derived from the LightGBM–SHAP framework, along with the ranked contributions of the top 25 features and their corresponding SHAP interpretations. All importance scores are computed using normalized mean absolute SHAP values. Collectively, these top 25 features account for 89.11% of the total normalized mean absolute SHAP importance, indicating that the key experiential features have been effectively and comprehensively captured. Overall, the Physical activity subdimension exhibited the highest relative importance (19.12%). Among these behavioral indicators, Walking emerged as the most influential feature, also ranking as the single most important variable across all predictors (11.73%), followed by Cycling (5.06%) and Running (2.33%). The Traffic Perception dimension ranked second, with Parking lot (7.90%) and Public transportation (6.45%) both contributing more than 5% to overall importance. Crowding perception ranked third, where both High crowd density (6.58%) and Low crowd density (3.32%) showed notable contributions. Other subdimensions, including Architectural landscape perception (7.35%), Environmental cleanliness perception (5.97%), Observational experience activity (5.76%), and Olfactory perception (5.13%), also exceeded 5% in relative importance. In contrast, Auditory perception (0%), Animal diversity perception (0.03%), and Spatial renewal cognition (0.51%) showed minimal relevance, each contributing less than 1% to the overall model.

Further SHAP dependence plots revealed the association patterns between individual indicators and SWB. Among all features, Walking not only showed the highest contribution but also had a strong positive contribution to model outputs. Similar positive associations were observed for Public transportation, Atmospheric freshness, Natural landscape appreciation, and Bridge. In contrast, several indicators showed negative associations, including High crowd density, Parking lot, and Environmental risk. Notably, features such as Cycling and Water quality exhibited context-dependent effects, with their association with SWB shifting between positive and negative depending on situational conditions.

4.2.2. Waterfront typologies feature importance

Fig. 6 and Fig. 7 illuminate the relative importance of features in

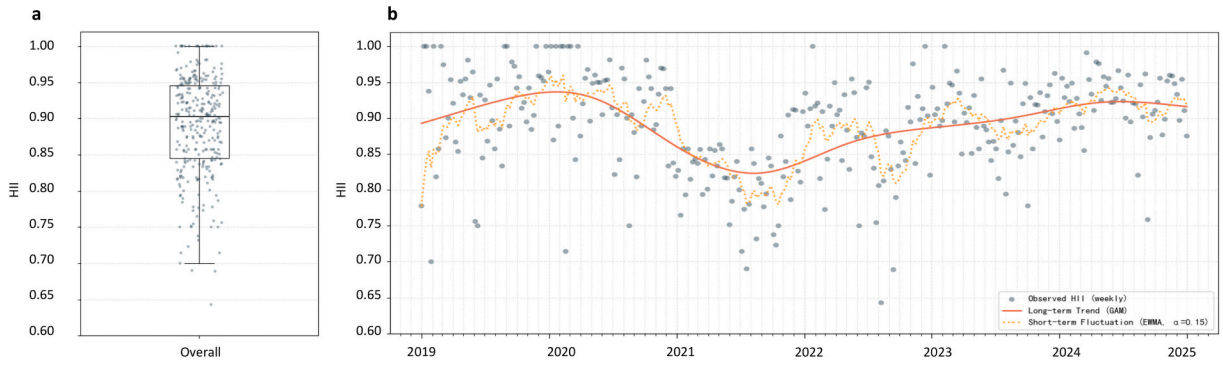


Fig. 3. (a) Overall HII distribution; (b) Temporal variation trend.

$$\Phi_{ij}(f,x)=\sum_{S\subseteq N\setminus\{i\}}\frac{|S|!(M-|S|-2)!}{2(M-1)!}\nabla_{ij}(S) \tag{8}$$

其中，φ_i与φ_{i_j}分别表示 Shapley 值与 Shapley 交互值；M 为特征总数，N 为特征集，S 为 N 中剔除特征 i 的子集。

4. 结果

4.1. 城市滨水区 HII 分布特征分析

4.1.1. 总体 HII 分布模式

当将所有不同类型滨水区的数据汇总后，总体HII值达到0.882，表明公众在滨水空间中普遍体验到较高的 SWB 水平。每周汇总则提供了关于时间动态的更细致洞察。如图3a所示，HII的中位数为0.90，四分位距（IQR）为0.85–0.95，最小值为0.70，这表明HII值始终集中在较高区间，仅偶尔出现下降。从时间趋势来看（图3b），2019–2020年间HII稳步上升，维持在约0.95的水平；随后在2020年至2022年间显著下降，于2021年中期降至最低点（0.80）；此后逐渐回升，自2023年起稳定在0.90以上。短期波动呈现出明显的季节性规律：HII在春季（3月至5月）略有上升或达到短期峰值；随后在夏季（6月至8月）显著下降至季节性低谷；秋季（9月至11月）再度回升；冬季（12月至次年2月）尽管略有下降，仍保持相对较高的水平。

4.1.2. 滨水建筑类型中的 HII 分布模式

图 4展示了不同滨水空间类型中 HII 的总体分布及时间变化趋势。滨河空间的中位 HII 值最高（0.92），分布相对集中（0.89–0.95），最小值为 0.80，表明这些空间中的公众幸福感总体保持在较高水平，仅出现小幅波动及偶发性下降。滨水公园的中位值略低（0.91），但分布最为集中（0.89–0.94）。

且最小值较高（0.86），表明其总体幸福指数水平略低于滨水区域，但时间稳定性更强。相比之下，滨水绿道区的中位数最低（0.90），第一四分位数（0.82）与最小值（0.70）也显著偏低，反映出整体幸福体验较弱且波动性更大（图4a）。从时间维度观察，滨水区域与公园空间均呈现相对稳定的长期趋势：HII值始终维持在0.90–0.93区间，短期波动有限，且总体呈交替上升态势。相较之下，绿道区则表现出最显著的波动性，其特征为反复经历衰退期与复苏期（图4b）。在季节性变化方面，滨水区域呈现明显的季节波动：HII值于春末至初夏（5月）达到峰值，夏末（7–8月）降至谷值（0.87），秋季逐渐回升，冬季（12–1月）再度小幅下降；而公园区域则最为稳定，多数月份HII值均介于0.90–0.93之间，季节性波动极小。然而，绿道则呈现出典型的“U型”季节性变化规律——年初（1月至2月）数值相对较高（约0.90），随后稳步下降至夏季最低值（8月，偶可低至0.75–0.80），继而于秋冬季节逐渐回升至约0.90（图4c）。

4.2. 基于 Shap 方法的相对特征重要性分析

4.2.1. 整体特征重要性

图 5展示了基于 LightGBM–Shap 框架从多个子维度得出的特征重要性结果，以及前 25 个特征的重要性排序及其对应的 Shap 解释值。所有重要性评分均采用标准化平均绝对 Shap 值计算得出。这 25 个关键特征共同贡献了总标准化平均绝对 Shap 重要性的 89.11%，表明核心体验特征已被有效且全面地捕捉。总体而言，“身体活动”子维度的相对重要性最高（19.12%）。在这些行为指标中，“步行”成为最具影响力的特征，同时也是所有预测变量中单项重要性最高的变量（11.73%），其次为“骑行”（5.06%）与“跑步”（2.33%）。“交通感知”维度位列第二，其中“停车场”（7.90%）与“公共交通”（6.45%）对总体重要性的贡献均超过 5%。“拥挤感知”维度排名第三，其中“高人群密度”（6.58%）与“低人群密度”（3.32%）均展现出显著贡献。其他子维度——包括建筑景观感知（7.35%）、环境清洁度感知（5.97%）、观察性体验活动（5.76%）以及嗅觉感知（5.13%）——在相对重要性方面也均超过5%。相比之下，听觉感知（0%）、动物多样性感知（0.03%）及空间更新认知（0.51%）的相关性极低，各自对整体模型的贡献度均低于1%。

进一步的形状依赖图揭示了各个指标与 SWB 之间的关联模式。在所有特征中，步行不仅贡献度最高，而且对模型输出也有显著的正向贡献。公共交通、空气清新度、自然景观欣赏度以及桥梁也呈现出类似的正向关联。相比之下，若干指标则显示出负向关联，包括高人群密度、停车场及环境风险。值得注意的是，骑行与水质等特征表现出情境依赖性效应，它们与 SWB 的关联会根据具体情境在正负之间转换。

4.2.2. 滨水类型学具有重要意义

图 6与图 7阐明了各特征在其中的相对重要性。

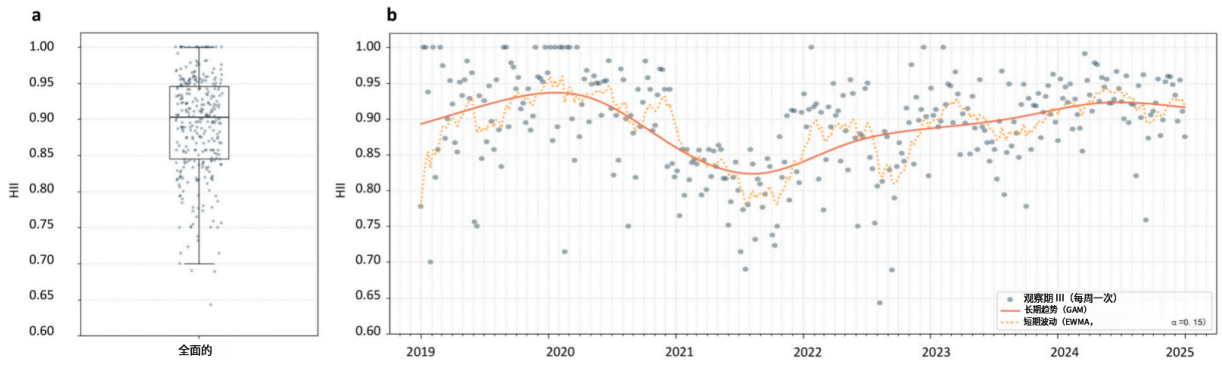


图 3.(a) HII 总体分布；(b) 时间变化趋势。

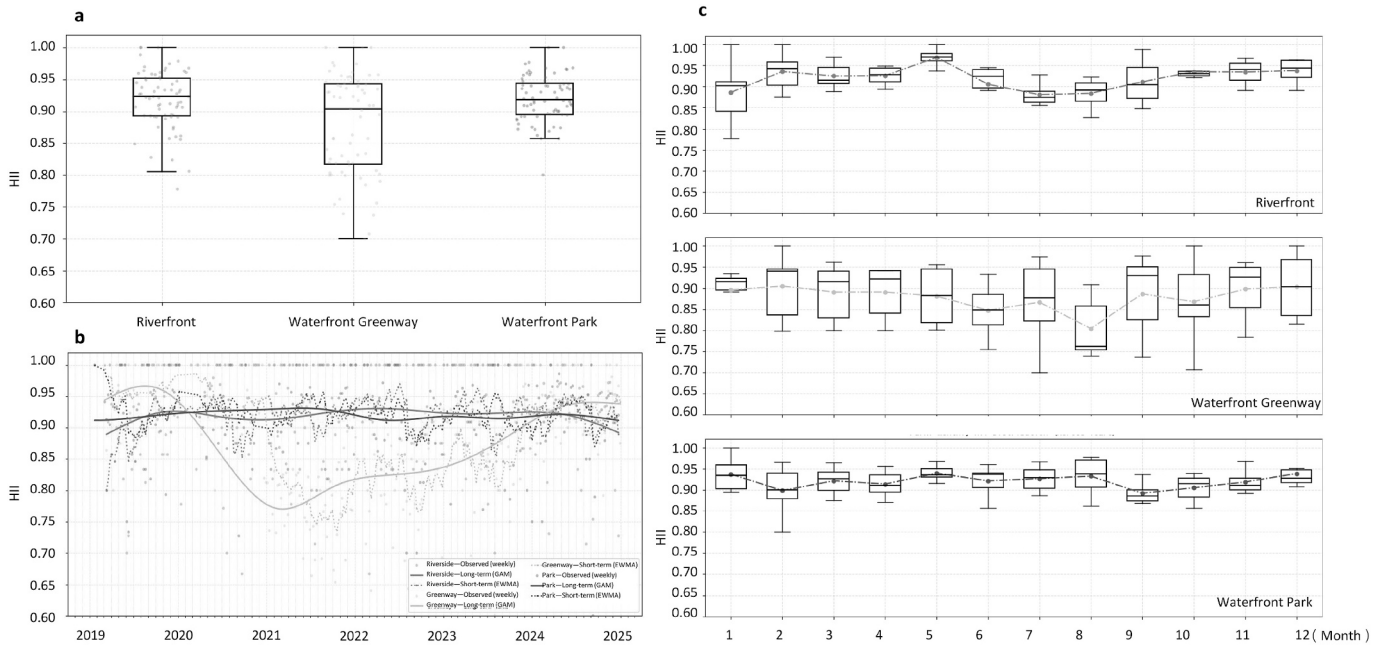


Fig. 4. (a) Overall HII distribution; (b) Long-term and short-term temporal trends; (c) Monthly variation patterns.

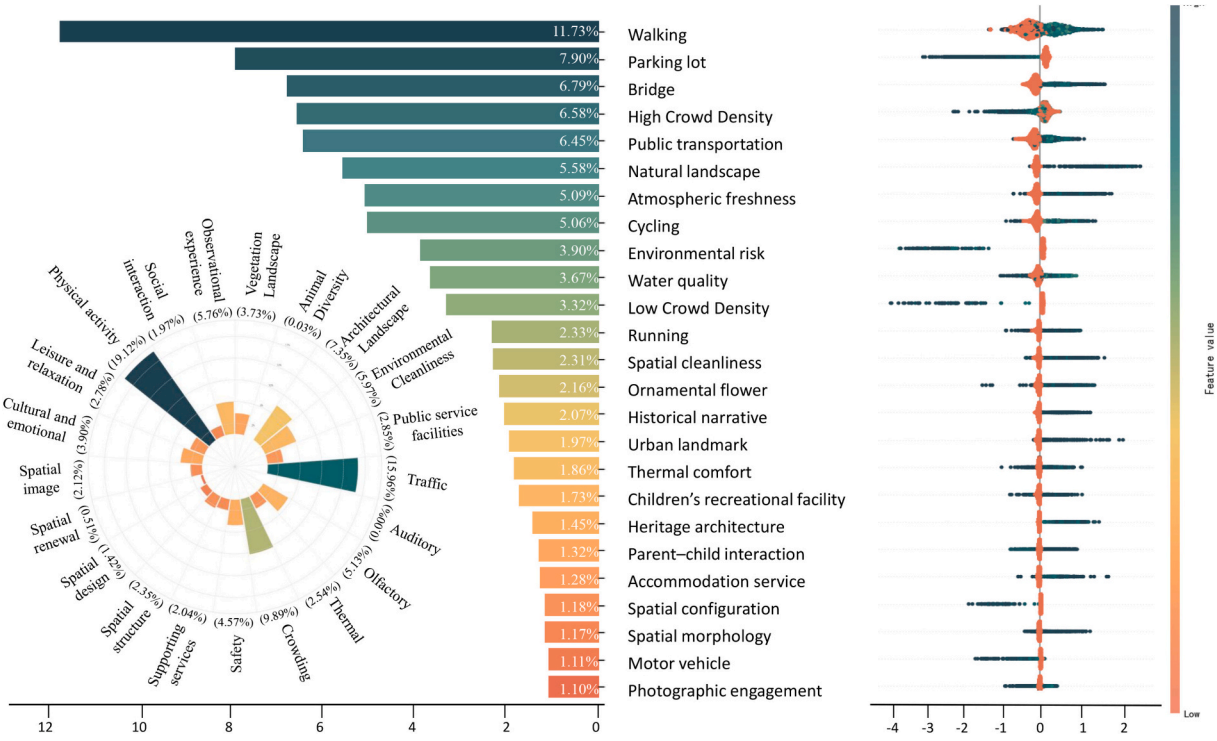


Fig. 5. SHAP-based feature importance for the overall sample.

waterfront settings at two analytical scales: the former compares commonalities and differences across riverfront, waterfront park, and waterfront greenway typologies, while the latter further details site-level patterns within each typology. Overall, Physical activity (e.g., Walking), Traffic Perception (e.g., Parking lot, Public transportation), and Crowding perception (e.g., High crowd density) consistently rank among the most influential features across typologies. Walking and Public transportation generally contribute positively to SWB, while Parking lot tends to contribute negatively. By contrast, High crowd density is more context-dependent: it is predominantly negative in greenways and parks, but on riverfronts can, under certain social or festive conditions, translate into a positive contribution to SWB.

Within the set of waterfront typologies examined, the SWB of riverfronts is strongly associated with Physical activity (15.90%), with Walking (12.11%) as the core contributor and Running (3.44%) also adding a positive effect. Beyond activity, Architectural landscape perception is salient across riverfronts, with Bridge exhibiting particularly high importance. Meanwhile, Observational experience activity (8.52%) and Leisure and relaxation activity (7.77%) also show strong positive associations. In addition, Cultural and emotional cognition,

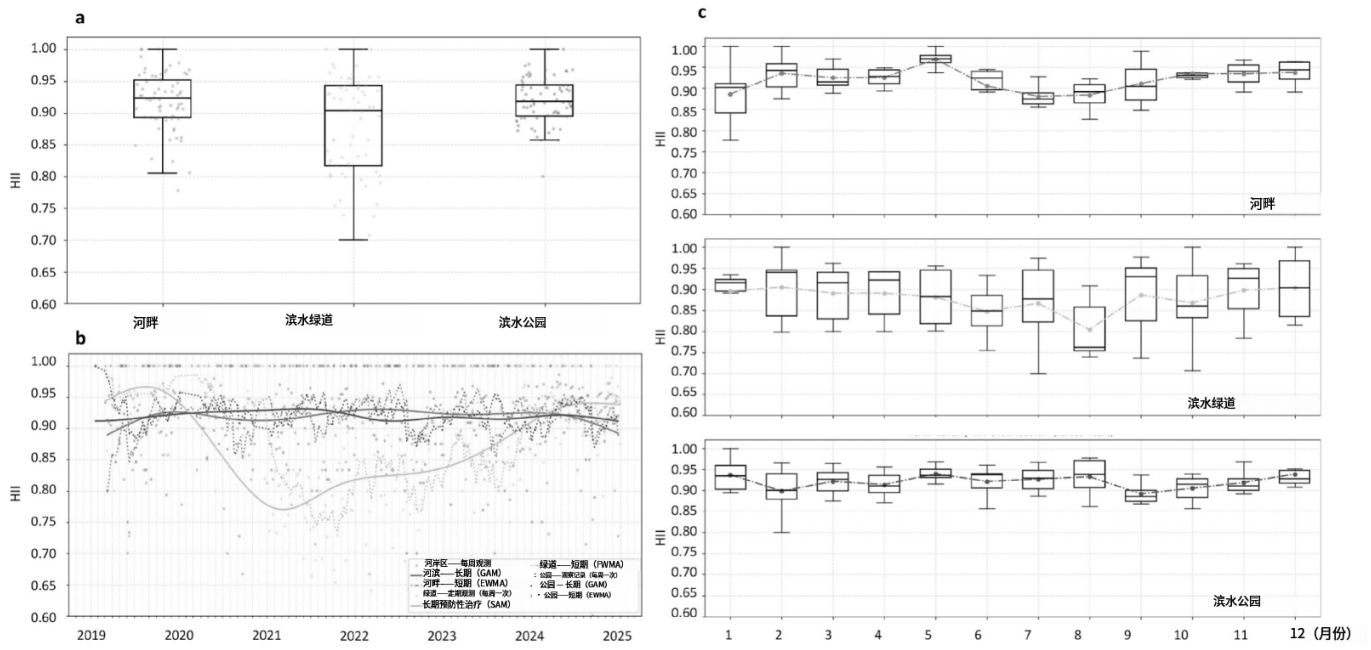


图 4.(a) HII 总体分布；(b) 长期与短期时间趋势；(c) 月度变化规律。

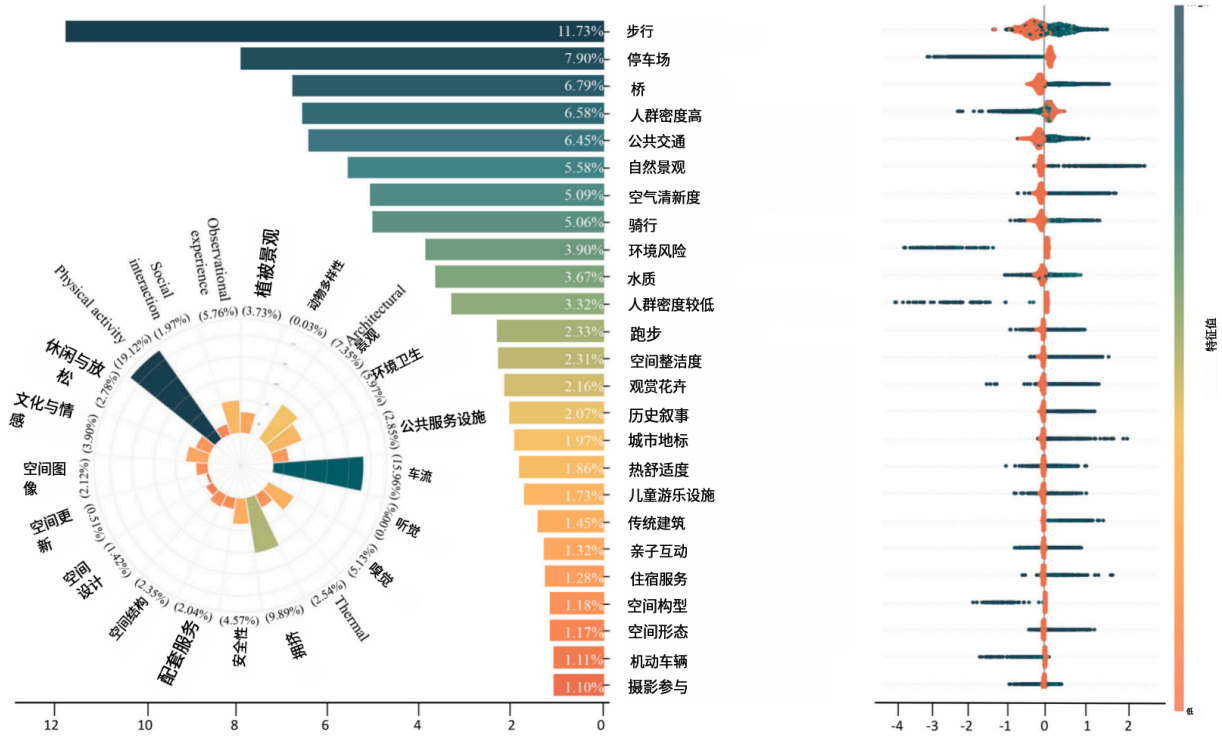


图 5.整体样本的基于形状的特征重要性分析。

从两个分析维度考察滨水环境：前者比较了河流滨水区、滨水公园及滨水绿道类型之间的共性与差异；后者则进一步细化每种类型内场地层面的特征模式。总体而言，体力活动（如步行）、交通感知（如停车场、公共交通）以及拥挤感知（如高人群密度）在各类型中始终位列最具影响力的特征之一。步行与公共交通通常对 SWB 产生积极影响，而停车场则多产生负面影响。相比之下，高人群密度的影响更具情境依赖性：在绿道与公园中，它主要产生负面影响；但在河流滨水区，在特定社会或节庆条件下，它却可能转化为对 SWB 的积极贡献。

在所考察的水滨类型中，河滨区域的 SWB 与身体活动密切相关（15.90%），其中步行（12.11%）是核心贡献因素，跑步（3.44%）也带来了积极影响。除活动外，建筑景观感知在水滨区域同样显著，其中桥梁的重要性尤为突出。同时，观察体验活动（8.52%）与休闲放松活动（7.77%）也显示出强烈的正相关。此外，文化与情感认知，如与河滨区

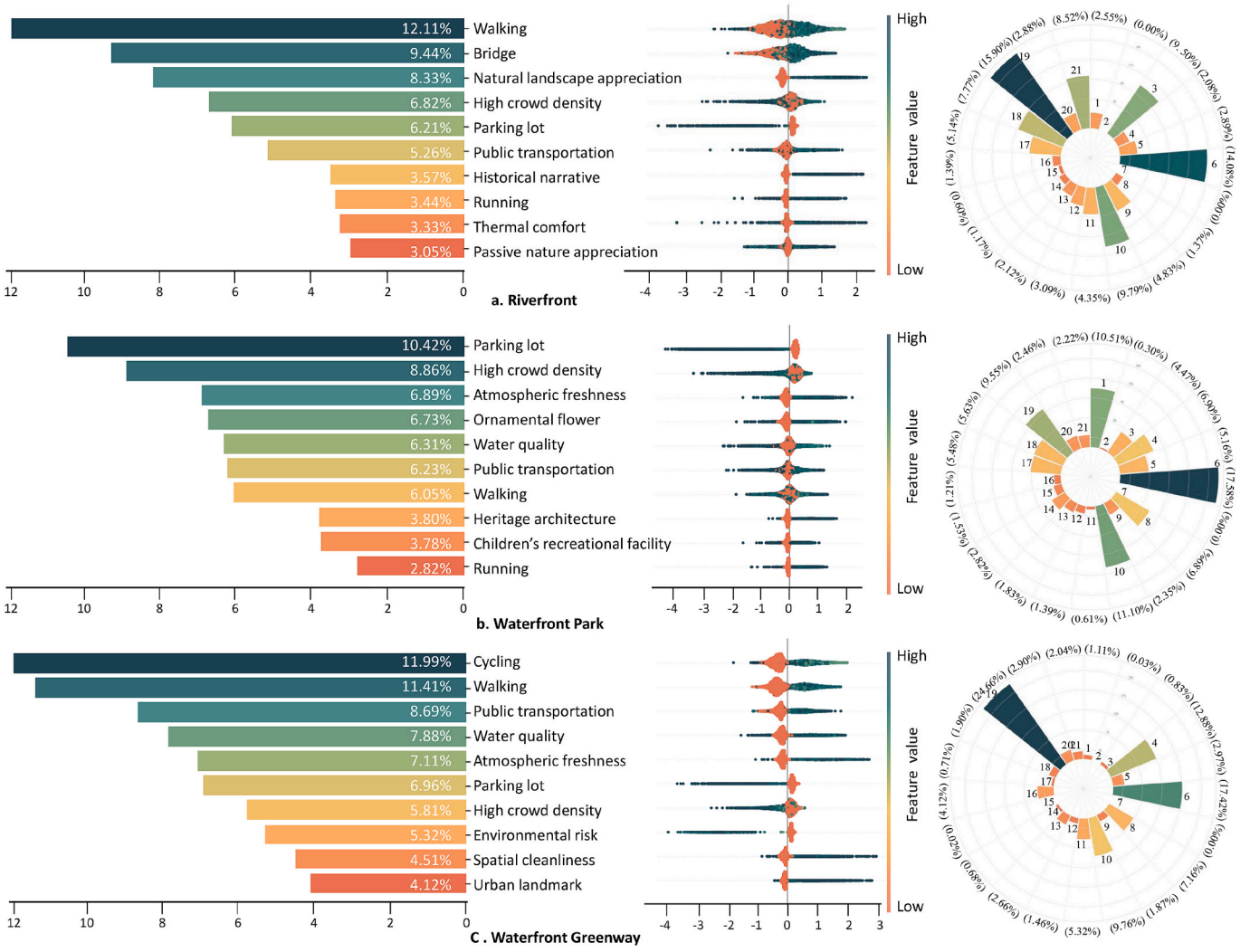


Fig. 6. SHAP-based feature importance across waterfront typologies. Note: The numbers in the radial charts correspond to the indicator IDs listed in Table 2.

such as historical narratives and emotional memory associated with the riverfront, holds considerable relative importance (5.14%). In waterfront parks, transportation perception (17.58%) was the factor most strongly associated with SWB, with Parking lot (10.42%) exhibiting the highest importance among all indicators, highlighting the critical role of accessibility in shaping park experiences. Perceived crowding ranked second (11.10%), among which High crowd density (8.86%) showed particularly high importance, suggesting that visitor activity intensity plays a significant role in well-being experiences. Meanwhile, natural environmental factors also contributed substantially to SWB, with Vegetation landscape perception (10.51%) emerging as one of the most influential indicators. In addition, Olfactory perception (6.59%) and Cultural and emotional cognition (5.48%) also demonstrated notable importance, indicating that well-being experiences in waterfront parks are shaped not only by environmental quality and accessibility, but also by natural perceptions and cultural experiences. The formation of SWB in waterfront greenways is strongly associated with physical activities (24.66%), whose relative importance is significantly higher than that of the other two types of spaces. Among these, Cycling (11.99%) and Walking (11.41%) are the dominant features, with cycling ranking first across all variables, which highlights the core function of greenways in providing high-quality spaces for physical exercise. Beyond common transportation and crowding related features, Environmental cleanliness perception also plays a notable role in greenways, where Water quality (7.88%) and Spatial cleanliness (4.51%) show strong associations. Similarly, olfactory perception, such as Atmospheric freshness (7.11%), shows a strong positive contribution. However, unlike the other two waterfront types, greenways also exhibit distinct negative features, particularly Safety cognition (e.g., Environmental risk), which is strongly associated with lower SWB.

Site-level analyses further revealed variations in the importance of key experiential factors within the same waterfront typology. Among the riverfronts, Wuchang Jiangtan was characterized by natural landscape appreciation, bridge imagery, and walking experiences, forming a composite experience centered on riverscape appreciation and recreational activities. Hanyang Jiangtan was distinguished by its prominent bridge landscapes, together with tranquil natural settings and cultural memory experiences, reflecting strong landscape and cultural characteristics. In contrast, Hankou Jiangtan exhibited stronger statistical associations with social ambiance and group interactive activities, with its experiential profile dominated by family-focused recreation and social leisure. Qingshan Jiangtan combined physical activity with landscape experiences, where walking, bridge scenery, landmarks, and photography-related activities jointly shaped its distinctive experiential profile. Waterfront parks exhibited more diverse well-being-related characteristics. Sha Lake Park, Northwest Lake Park, and Ziyang Lake Park were primarily associated with vegetation landscapes, flower appreciation, and family-oriented activities, highlighting their functions as spaces for nature appreciation and social interaction. Simeitang Park, Huanzi Lake Park, and Moon Lake Park integrated historic architecture,

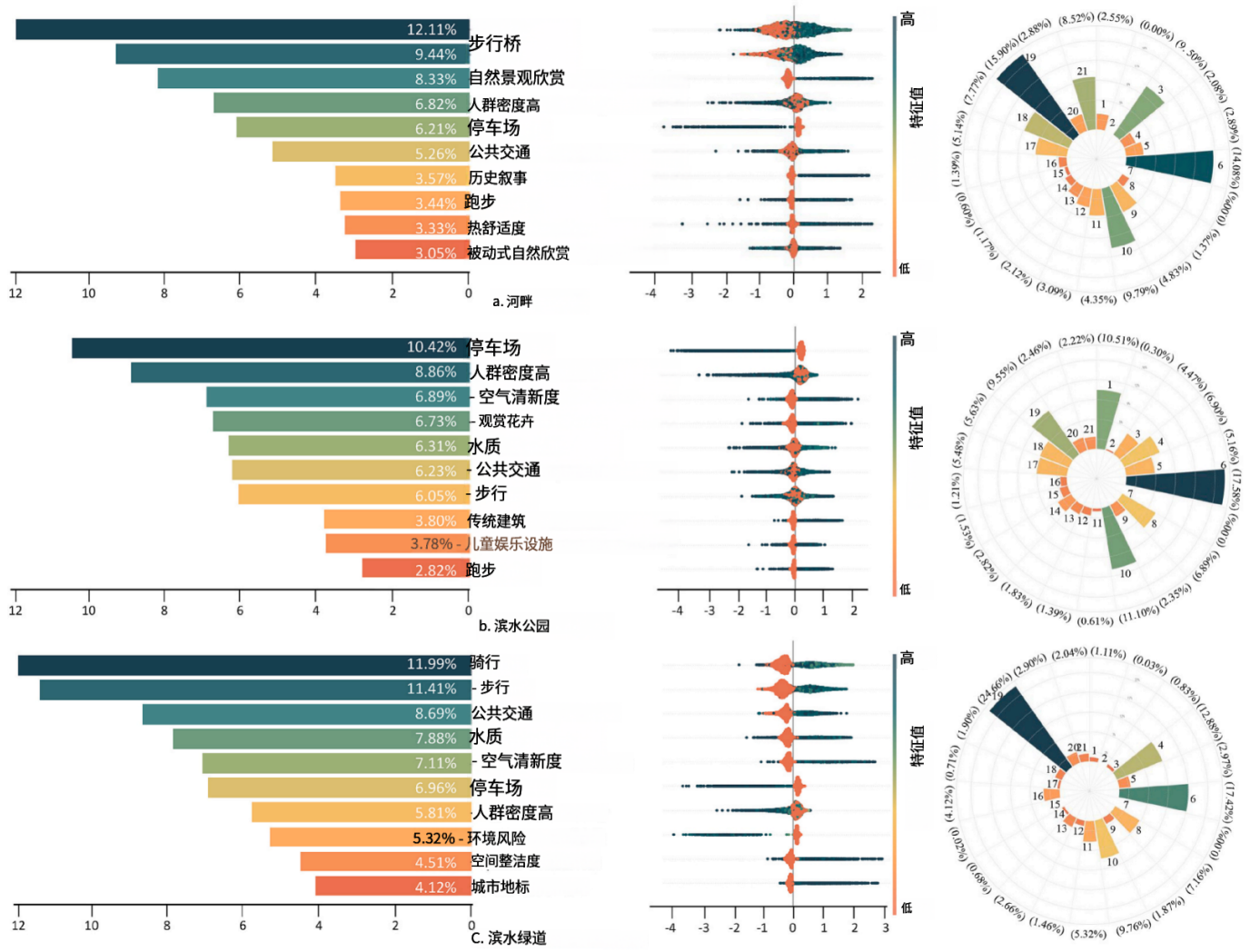


图 6。不同滨水类型下的基于形状的特征重要性。注：径向图表中的数值对应于表 2中列出的指标 ID。

域相关的历史叙事和情感记忆，也占有相当重要的相对权重（5.14%）。在水滨公园中，交通感知（17.58%）是与 SWB 关联最紧密的因素，而停车场（10.42%）在所有指标中重要性最高，凸显了可达性对塑造公园体验的关键作用。感知拥挤度位居第二（11.10%），其中高人群密度（8.86%）尤为重要，表明游客活动强度对福祉体验具有重要影响。与此同时，自然环境因素也对 SWB 产生了显著影响，其中植被景观感知（10.51%）成为最具影响力的指标之一。此外，嗅觉感知（6.59%）以及文化与情感认知（5.48%）也显示出重要意义，这表明滨水公园中的福祉体验不仅受环境质量与可达性的影响，还受到自然感知与文化体验的共同塑造。滨水绿道中 SWB 的形成与体力活动密切相关（24.66%），其相对重要性显著高于其他两类空间。其中，骑行（11.99%）与步行（11.41%）是主要特征，骑行在所有变量中位居首位，这凸显了绿道在提供高质量体力活动空间方面的核心功能。除常见的交通与拥挤相关特征外，环境清洁感知在绿道中也发挥着重要作用，其中水质（7.88%）与空间清洁度（4.51%）均显示出强相关性。

同样地，嗅觉感知，如空气清新度（7.11%），也显示出强烈的正向影响。然而，与其他两种滨水类型不同，绿道还表现出一些明显的负面特征，尤其是安全性认知（例如环境风险），这与较低的 SWB 值密切相关。

场地层面的分析进一步揭示了同一滨水类型区内关键体验要素的重要性存在差异。在各类滨水区域中，武昌江滩以自然景观欣赏、桥梁意象及步行体验为特色，形成了以河景欣赏与休闲活动为核心的综合体验；汉阳江滩则以鲜明的桥梁景观、静谧的自然环境及文化记忆体验为显著特征，展现出强烈的景观与文化特质；相比之下，汉口江滩与社会氛围及群体互动活动呈现更强的统计学关联，其体验特征以家庭导向型休闲及社交休闲活动为主；青山江滩则将体力活动与景观体验相结合，步行、桥梁景致、地标建筑及摄影相关活动共同塑造了其独特的体验特征；滨水公园则展现出更为多样化的健康福祉相关特征。沙湖公园、西北湖公园及紫阳湖公园主要与植被景观、花卉观赏及家庭友好型活动相关联，凸显其作为自然欣赏与社会互动空间的功能定位。西梅塘公园、

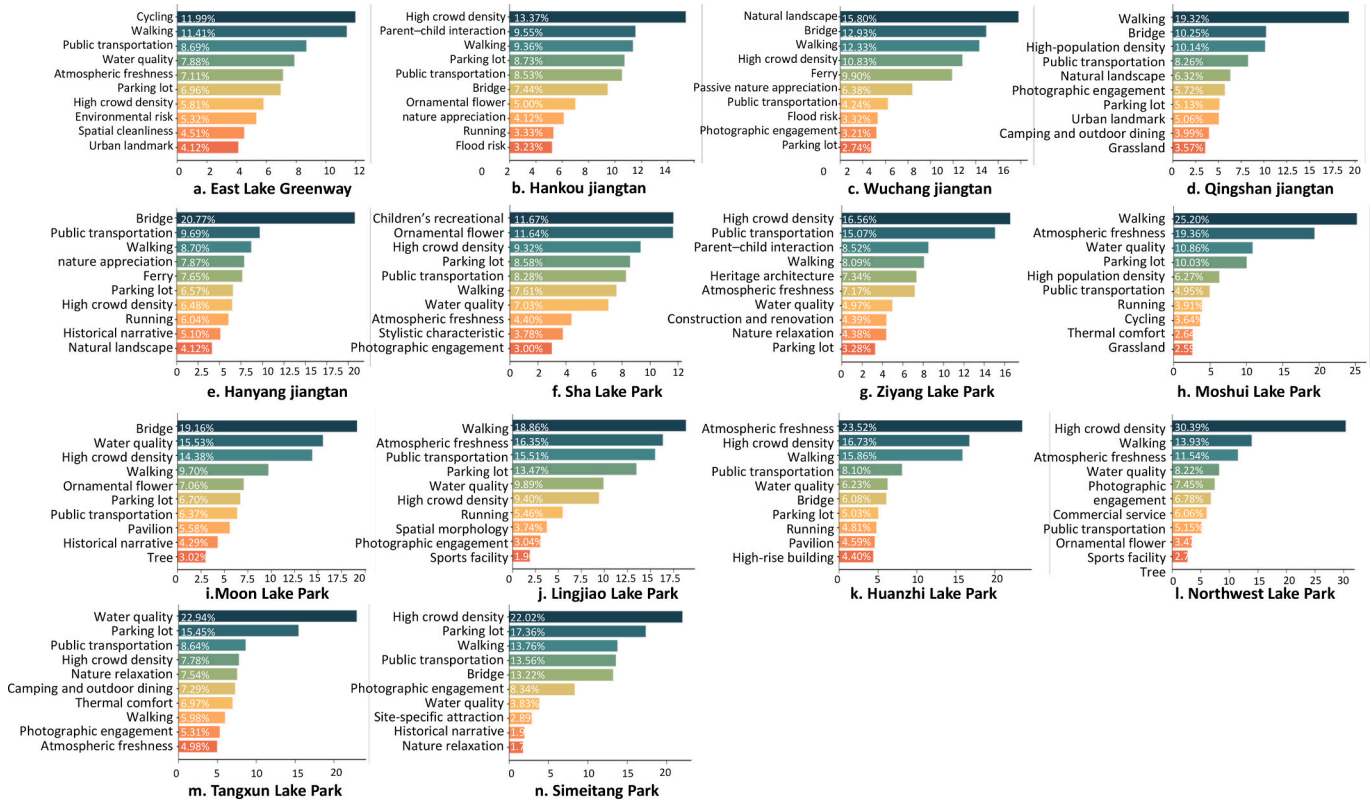


Fig. 7. SHAP-based feature importance at the site level.

cultural narratives, and landscape imagery, demonstrating strong cultural and historical value. Lingjiao Lake Park and Moshui Lake Park were characterized by jogging, cycling, and sports facilities, forming multifunctional spaces that combine physical exercise with natural landscape experiences. Tangxun Lake Park presented prominent associations with camping, dining and thermal comfort perceptions, corresponding to its suitability for long-duration visits and seasonal leisure activities.

4.2.3. *Nonlinear effects and feature interactions*

The associations between different features and SWB are often not purely linear but exhibit complex nonlinear variations and disparities. Fig. 8 illustrates the response patterns of the top eight features, showing how their associations with SWB vary across different intensity levels. The results reveal substantial differences in the relationships between feature attention and SHAP values. Features such as Bridge and High crowd density generally exhibit a linear relationship, where the association with happiness consistently increases or decreases with greater attention, showing distinct and stable trends. In contrast, Walking, Parking Lot, Cycling, and Natural Landscape exhibited pronounced nonlinear response patterns. Specifically, Walking reached its maximum positive contribution at an attention level of approximately 0.58, after which its positive effect gradually diminished, indicating a clear pattern of diminishing marginal returns. Parking Lot showed a distinct turning point around 0.57, beyond which its negative influence tended to stabilize. Cycling, Natural Landscape, and Atmospheric Freshness exhibited

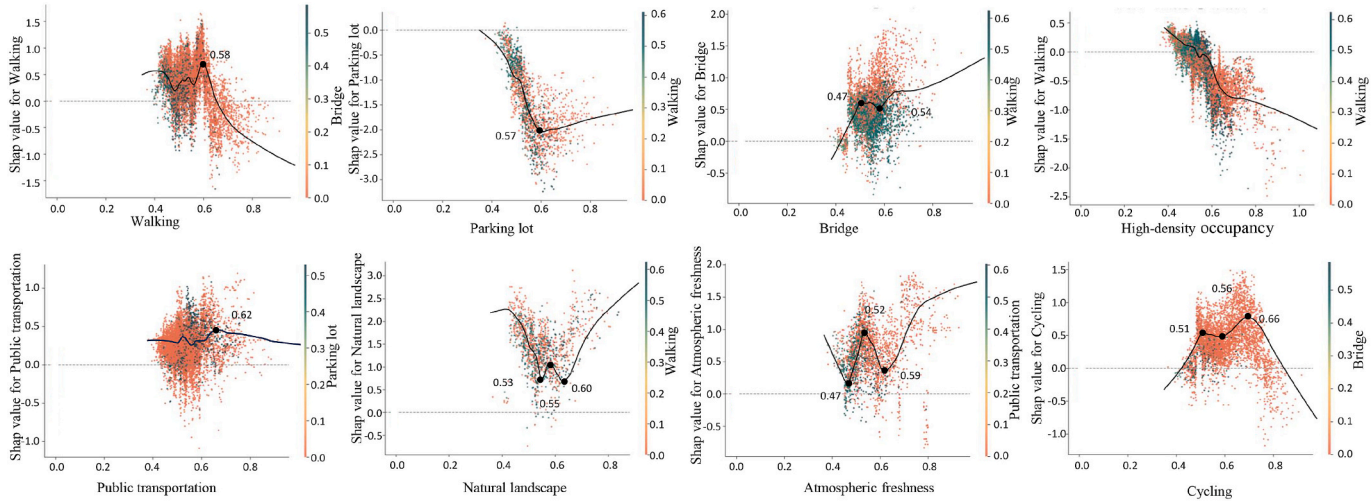


Fig. 8. Relationships and interaction patterns of key features.

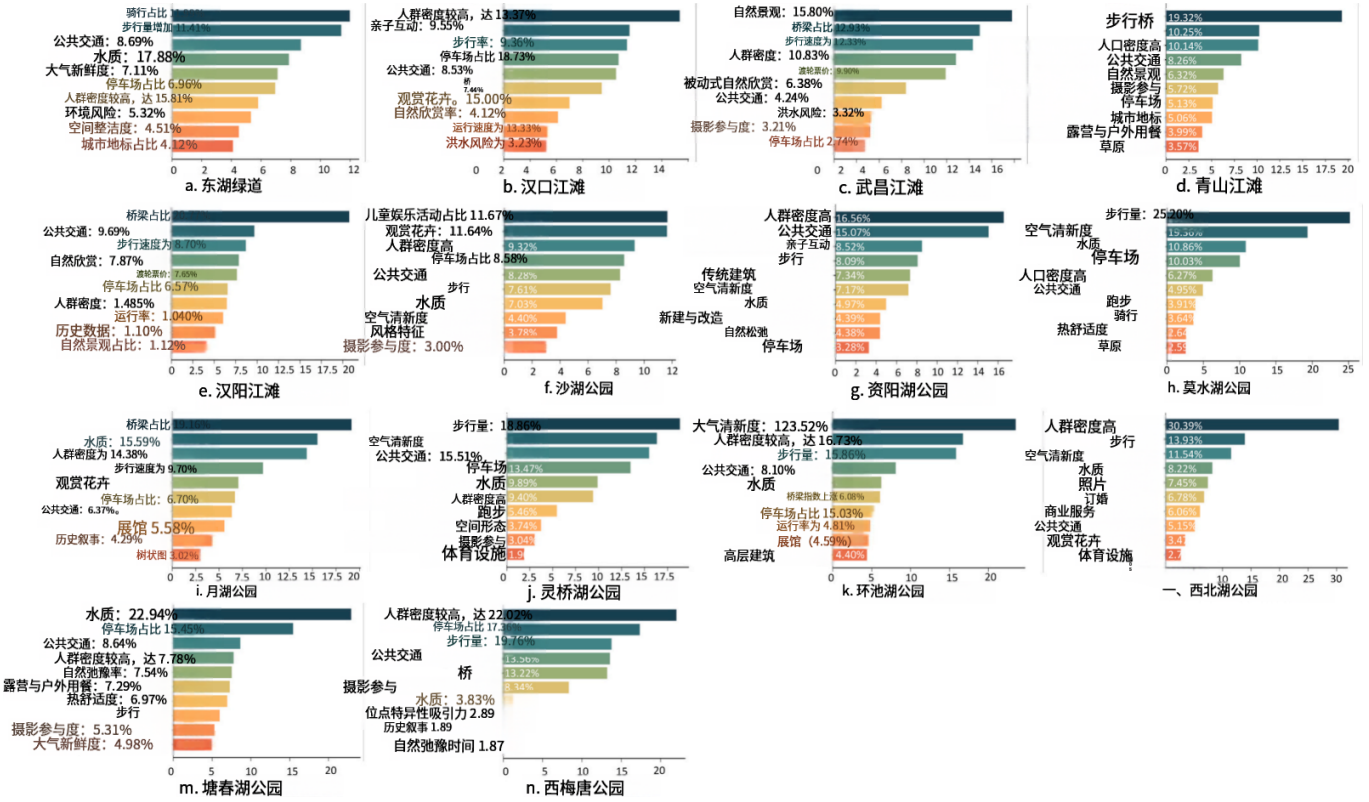


图 7.站点层面的基于形状的特征重要性。

环子湖公园及月湖公园则融合了历史建筑、文化叙事与景观意象，展现出深厚的文化与历史价值。灵桥湖公园与莫水湖公园以慢跑、骑行及体育设施为特色，打造出将体育锻炼与自然景观体验相结合的复合型空间。唐逊湖公园则与露营、餐饮及温泉舒适度感知密切相关，这与其适合长期游览及季节性休闲活动的特性相契合。

4.2.3. 非线性效应与特征交互作用

不同特征与 SWB 之间的关联通常并非纯粹的线性关系，而是呈现出复杂的非线性变化与差异。图8展示了前八个特征的响应模式，揭示了这些特征与 SWB 的关联在不同强度水平下的变化情况。研究结果揭示了特征关注度与Shap值之间的关系存在显著差异。诸如“桥梁”和“高人群密度”等特征通常表现出线性关系，即随着关注度的增加，与幸福感的关联持续增强或减弱，呈现出清晰且稳定的趋势。相比之下，“步行”、“停车场”、“骑行”及“自然景观”则表现出明显的非线性响应模式。具体而言，“步行”在关注度约为0.58时达到最大正向贡献，此后其正向效应逐渐减弱，显示出边际收益递减的明显趋势；而“停车场”则在关注度约为0.57时出现一个明显的转折点，此后其负面影响趋于稳定。骑行、自然景观与空气清新度均呈现出多个关键转折点，表明这些因素对 SWB 的影响具有阶段依赖性。这些发现表明，这些体验因素

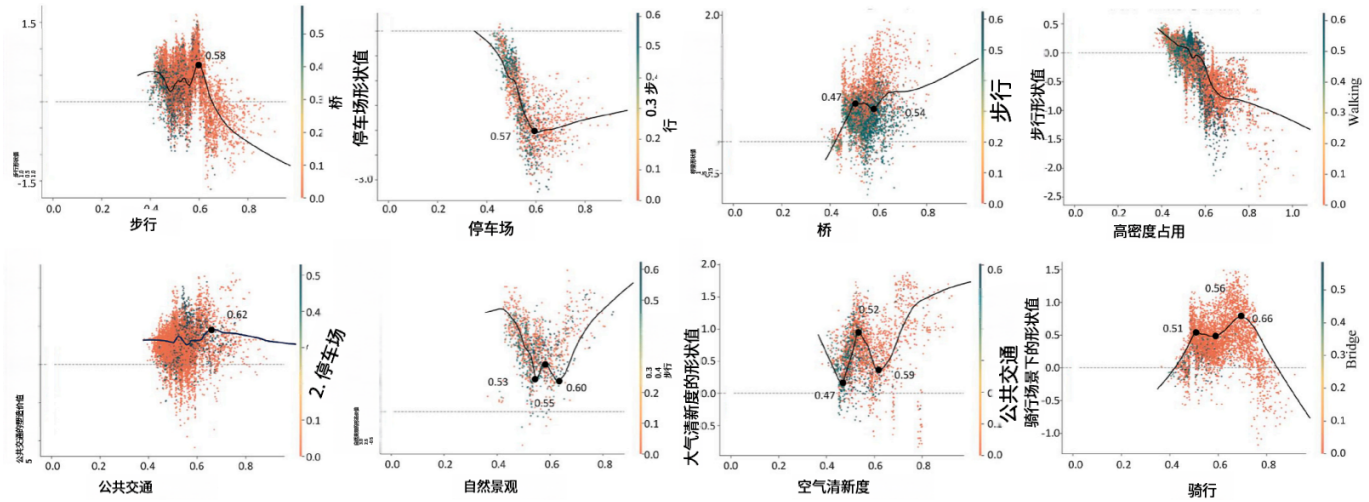


图 8.关键特征之间的关联性与交互模式。

multiple critical change points, suggesting stage-dependent effects on SWB. These findings indicate that the contributions of these experiential factors to well-being do not increase indefinitely with higher attention levels, but are instead constrained by contextual conditions and experiential thresholds.

In addition, a single text often encompasses multiple features whose complex interactions are associated with its emotional outcome. Fig. 8 further illustrates the interaction patterns among the top eight core features and their strongest interacting counterparts. The results indicate several positive synergistic relationships. For instance, the presence of Bridge in waterfront spaces enhances the positive contribution of Walking to SWB. Meanwhile, negative interactions are also observed; for example, strong Public transportation tends to weaken the positive contribution of Atmospheric freshness to SWB. These findings suggest that different combinations of environmental and behavioral factors are associated with differentiated emotional outcomes, highlighting the context-dependent nature of waterfront experiences.

To further explore potential heterogeneity across different waterfront typologies, grouped SHAP dependence analyses were conducted for riverfront, waterfront parks, and waterfront greenways. The results generally supported the findings from the full-sample analysis, with the response trends of key indicators remaining largely consistent across typologies, suggesting the robustness of the associations between major experiential factors and SWB. Nevertheless, differences were observed in the primary interaction variables of several indicators across waterfront types. For example, walking primarily interacted with bridge perception in riverfronts, whereas it showed a stronger interaction with High crowd density in waterfront greenways. These findings suggest that although the overall association patterns of core experiential factors remain relatively stable, their specific interaction patterns are context-dependent and influenced by the spatial characteristics and functional attributes of different waterfront environments (see Fig. S2 in Appendix H).

5. Discussion

5.1. HII patterns and feature importance in waterfront spaces

From a temporal perspective, variations in waterfront HII are jointly associated with social events and climatic conditions. The long-term trend shows that the decline from 2020 to 2022 was likely associated with reduced spatial use caused by mobility restrictions during the COVID-19 pandemic (X. Wu et al., 2023), whereas the rapid rebound after 2023 reflects the recovery of the happiness effect of waterfront spaces following the restoration of social order. This rebound vividly demonstrates the role of waterfronts as vital public environments that facilitated the restoration of residents' SWB and everyday normality during post-crisis recovery, representing an important manifestation of urban resilience (Grace et al., 2024). Short-term fluctuations exhibit a seasonal pattern: the waterfront happiness index tends to decline under extremely high or low temperatures, which to some extent suggests a correlation between thermal conditions and the waterfront happiness index. Research has shown that weather factors such as sunlight, temperature, and precipitation can indirectly influence happiness by affecting individuals' physiological states, emotional experiences, and social interactions (X. Wu et al., 2023). Despite these temporal fluctuations, the overall HII of waterfront spaces has remained consistently high over the long term, highlighting their enduring role in emotional regulation and psychological support within urban environments.

Physical activity was identified as the experiential factor most strongly associated with SWB in waterfront spaces, and this key finding is consistent with the prevailing conclusions of existing research. Previous studies have generally demonstrated that waterfront environments provide residents with opportunities for walking, running, cycling, and other forms of physical activity, while higher levels of activity participation are typically associated with better health outcomes and higher SWB levels (Bell et al., 2021; Murrin et al., 2023). However, previous research has rarely systematically compared the relative contributions of physical activity with other multidimensional experiential factors, including perceptual, cognitive, and behavioral dimensions. This study further reveals that, among all identified experiential factors, physical activity exhibits the strongest association with SWB, highlighting behavioral experience as a key experiential mechanism underlying the association between waterfront environments and well-being. This finding suggests that behavior is not merely an external manifestation of individual states but also constitutes an essential component in the formation of positive well-being experiences (Y. Liu et al., 2019; Murrin et al., 2023). From the perspective of Person–Environment Fit Theory, waterfront spaces typically provide long, continuous activity corridors and open visual environments, which better satisfy residents' needs for exercise and recreational activities (Caplan and Van Harrison, 1993; French, 1974). Such a high degree of compatibility between environmental resources and individual needs facilitates physical activity engagement and generates comfortable and pleasurable experiences, representing an important advantage of waterfront spaces compared with other urban open spaces (Benton et al., 2021). This person–environment fit not only promotes participation in physical activities but also enhances individuals' positive experiences during activities, thereby contributing to higher levels of SWB. Meanwhile, behaviors related to relaxation, social interaction, and aesthetic appreciation also showed substantial contributions, confirming the multifunctional and composite health value of waterfront spaces beyond their role as simple exercise venues. This finding further expands current understandings of the diverse pathways through which blue spaces support human well-being.

Moreover, beyond behavioral patterns, perceptual features of the built and social environments are also strongly associated with people's happiness experiences in waterfront spaces. Among these factors, Traffic Perception and Crowding Perception exhibited the greatest contributions. Previous studies have primarily focused on objective accessibility indicators, such as road-network density and spatial distance, to examine the positive effects of accessibility on waterfront use frequency and health benefits (Tieges et al., 2022; Zhan et al., 2026). In contrast, this study uses Traffic Perception to capture individuals' experiences of transportation convenience and accessibility. This finding extends existing accessibility research by incorporating the human-centered experiential dimension and provides a more comprehensive understanding of how accessibility contributes to well-being benefits in waterfront environments. Regarding Crowding Perception, existing studies have mainly focused on external built-environment characteristics and internal spatial quality, while the impacts of population density and subjective crowding experiences on well-being remain relatively underexplored (H. Cao et al., 2024; Gao and Fang, 2025; Garrett et al., 2023; G. Zhu et al., 2023). This study further reveals a bidirectional and nonlinear relationship between perceived crowding and SWB, suggesting that the effects of crowding are highly context-dependent. This phenomenon can be interpreted through the lens of Environmental Load Theory. Individuals are continuously exposed to environmental stimuli, and their psychological responses depend largely on the intensity of such stimulation. Moderate levels of environmental stimulation can enhance spatial vitality and social engagement, thereby generating positive emotional experiences, whereas excessive stimulation may exceed individuals' coping capacity and induce stress, fatigue, and cognitive overload (Milgram, 1970; Stokols, 1972). Therefore, high population density does not necessarily result in negative experiences. In contexts such as festivals, public events, and social gatherings, moderate crowd concentration may enhance place vitality and strengthen the social atmosphere, thereby promoting positive emotions. However, excessive crowding may increase psychological burden and reduce SWB. This provides a possible explanation for the bidirectional association between crowding and SWB observed in this study. Notably, this study found that soundscape had limited independent explanatory

对幸福感的贡献并不会随着注意水平的提高而无限增加，而是受到环境条件和体验阈值的限制。

此外，单一文本往往包含多个特征，这些特征之间复杂的相互作用与其情感结果密切相关。图8进一步展示了前八个核心特征及其最强交互对应特征之间的相互作用模式。研究结果揭示了若干积极的协同效应。例如，滨水空间中“桥梁”的存在可增强“步行”对SWB的积极贡献。与此同时，也观察到负向交互作用；例如，强烈的“公共交通”往往会削弱“空气清新度”对SWB的积极贡献。这些发现表明，环境因素与行为因素的不同组合与不同的情感结果相关，凸显了滨水体验的语境依赖性。

为进一步探究不同滨水类型间的潜在异质性，本研究对滨水区、滨水公园及滨水绿道分别进行了分组Shap依赖分析。结果总体上支持全样本分析的结果，关键指标的响应趋势在不同类型中基本保持一致，这表明主要体验因素与SWB之间的关联具有稳健性。然而，在若干指标的主要交互变量方面，不同滨水类型间仍存在差异。例如，在滨水区，步行主要与桥梁感知产生交互作用；而在滨水绿道中，步行则与高人群密度表现出更强的交互作用。这些发现表明，尽管核心体验因素的总体关联模式相对稳定，但其具体交互模式却取决于环境背景，并受不同滨水环境的空间特征与功能属性影响（参见附录H中的图S2）。

5. 讨论

5.1. 滨水空间中的HII模式与特征重要性

从时间维度来看，滨水HII的变化与社会事件及气候条件密切相关。长期趋势显示，2020年至2022年间滨水HII的下降，可能与新冠疫情期间的出行限制导致的空间使用减少有关（X.Wu等，2023）；而2023年后滨水HII的迅速回升，则反映了社会秩序恢复后滨水空间幸福效应的复苏。这一回升生动地证明了滨水区作为重要公共环境的作用——在危机后恢复期，它助力居民重建SWB与日常生活常态，体现了城市韧性的重要表现（Grace 等，2024）。短期波动则呈现季节性特征：在极端高温或低温条件下，滨水幸福指数往往下降，这在一定程度上表明温度条件与滨水幸福指数之间存在关联。研究表明，阳光、温度和降水等天气因素可通过影响个体的生理状态、情绪体验及社会互动，间接影响幸福感（Wu X. 等，2023）。尽管存在这些时间性波动，滨水空间的整体HII值在长期内仍持续保持较高水平，这凸显了滨水空间在城市环境中对情绪调节与心理支持所发挥的持久作用。

体力活动被确定为与滨水空间中的SWB关联最密切的体验性因素，这一关键发现与现有研究的普遍结论相一致。既往研究普遍表明，滨水环境为居民提供了步行、跑步、骑行及其他形式的体力活动机会，而较高的活动参与率通常与更好的健康结局及更高的SWB水平相关（Bell 等，2021；Murrin 等，2023）。然而，既往研究很少系统性地比较体力活动与其他多维体验性因素（包括感知、认知及行为维度）的相对贡献。本研究进一步揭示，在所有已识别的体验性因素中，体力活动与SWB的关联最为显著，这凸显了行为体验作为滨水环境与幸福感之间关联的关键体验机制。这一发现表明，行为不仅是个体状态的外在表现，更是构建积极幸福感体验的关键组成部分（刘等人，2019；Murrin等人，2023）。从“人-环境适配理论”的角度来看，滨水空间通常提供漫长且连续的活动廊道以及开阔的视觉环境，从而更好地满足居民对锻炼与休闲活动的需求（Caplan与Van Harrison，1993；French，1974）。环境资源与个体需求之间的高度契合，有助于促进身体活动的参与，并营造出舒适愉悦的体验，这相较于其他城市开放空间而言，是滨水空间的重要优势（Benton 等人，2021）。这种人-环境适配不仅推动居民参与身体活动，还能增强个体在活动过程中的积极体验，进而有助于提升SWB水平。与此同时，与放松、社交互动及审美欣赏相关的行为也显示出显著贡献，这证实了滨水空间除作为单纯运动场所外，还具有多功能性与综合健康价值。该发现进一步拓展了当前对蓝色空间促进人类福祉之多元作用路径的认知。

此外，除了行为模式之外，建成环境与社会环境的感知特征也与人们在滨水空间中的幸福感体验密切相关。在这些因素中，交通感知与拥挤感知对幸福感的贡献最为显著。既往研究主要聚焦于客观可达性指标，如路网密度与空间距离，以探讨可达性对滨水空间使用频率及健康效益的积极影响（Tieges 等，2022；Zhan 等，2026）。相比之下，本研究采用“交通感知”来捕捉个体对交通便利性与可达性的体验。这一发现通过引入以人为本的体验维度，拓展了现有可达性研究，从而更全面地理解可达性如何促进滨水环境中的福祉效益。关于拥挤感知，现有研究主要聚焦于外部建成环境特征与内部空间品质，而人口密度及主观拥挤体验对幸福感的影响仍相对缺乏探讨（Cao 等，2024；Gao 与 Fang，2025；Garrett 等，2023；Zhu 等，2023）。本研究进一步揭示了感知拥挤度与SWB之间存在双向且非线性的关系，表明拥挤效应具有高度情境依赖性。这一现象可通过“环境负荷理论”进行阐释：个体持续暴露于环境刺激中，其心理反应在很大程度上取决于此类刺激的强度。适度的环境刺激可增强空间活力与社会互动，进而引发积极情绪体验；而过度的刺激则可能超出个体的应对能力，导致压力、疲劳及认知超载（Milgram，1970；Stokols，1972）。因此，高人口密度并不必然导致负面体验。在节日、公共活动及社交聚会等场景中，适度的人群聚集可增强场所活力，强化社交氛围，从而促进积极情绪的产生。然而，过度拥挤则可能增加心理负担，降低SWB值。这为本研究中观察到的拥挤与SWB之间的双向关联提供了可能的解释。值得注意的是，本研究发现声景对SWB的解释力有限，这与既往强调声景特征在塑造环境体验与幸福

power for SWB, which differs from previous studies highlighting the important role of soundscape characteristics in shaping environmental experiences and well-being (F. Liu et al., 2022; J. Zhang et al., 2025). This discrepancy may be attributed to the representational characteristics of social media text data. Compared with visual landscapes and activity-related experiences, soundscape experiences are more transient, context-dependent, and less explicitly expressed in textual narratives. Consequently, soundscape-related experiences may be underrepresented in user-generated content.

Further analysis revealed substantial interaction patterns among waterfront features in relation to SWB. Some features exhibited positive synergistic associations—for example, a positive interaction between Bridge and Walking was observed in their contribution to SWB. Bridges not only increase the visual appeal of the environment but also provide extended pathways and visual focal points, thereby enriching the walking experience (Plasencia-Lozano, 2015). In contrast, public transportation showed a negative interaction with Atmospheric freshness in their contribution to SWB, likely because frequent traffic flow and associated emissions negatively affect people's subjective perception of air quality. These findings indicate that SWB is associated with multiple features rather than a single factor, and their relationships vary under specific contextual conditions.

5.2. SWB differences and association patterns across waterfront typologies

Different types of waterfront spaces exhibited substantial variations in both HII levels and temporal dynamics, which were closely associated with differences in their experiential characteristics. Previous studies have typically classified waterfront spaces according to waterbody attributes, such as rivers, lakes, and canals, and compared the associations between different waterbody types and health-related outcomes (Gao and Fang, 2025; F. Liu et al., 2022; Luo et al., 2023; McDougall et al., 2022). This study extends existing research by advancing both the classification paradigm and theoretical interpretation of waterfront–health relationships. Specifically, it not only identifies variations in well-being levels and seasonal dynamics across different waterfront space types but also further explores the potential explanations for these differences.

Specifically, riverfronts demonstrate the highest overall HII level, accompanied by moderate seasonal fluctuations, which are associated with a diversified behavioral structure—particularly the combination of physical activities and recreational leisure—linked to positive emotions and psychological restoration. Broad and continuous waterfront trails, together with unobstructed spatial layouts, are strongly associated with physical activity and higher levels of positive emotions and mental vitality (Yan et al., 2024). Meanwhile, leisure activities such as enjoying the river breeze or appreciating the scenery create typical scenes of emotional restoration and sensory pleasure, allowing visitors to relax and alleviate mental fatigue and cognitive load through visual and auditory stimulation (Severin et al., 2022; G. Zhu et al., 2023). The architectural and landscape features of riverfronts, particularly the iconic Yangtze River Bridge, together with the expansive river views and the surrounding riverfront buildings, form a richly layered visual interface. This landscape configuration not only enriches the exercise experience by enhancing the visual appeal of the space but also enhances the waterfront's recreational appeal by reinforcing the city's image and local identity (Lynch, 2023).

The overall HII level of waterfront parks is slightly lower than that of riverfronts, yet it remains the most stable throughout the year, showing the weakest sensitivity to seasonal and climatic variations. Transport accessibility and environmental quality show strong combined associations with SWB in waterfront parks. Compared with riverfronts and greenways, waterfront parks are typically embedded within residential areas on a micro-scale and spatially dispersed pattern, making them the most frequently accessed waterfront environments for daily visits. Therefore, Traffic Perception is the most strongly associated factor with SWB in waterfront parks. The higher the level of transport accessibility, the more frequently and intensively residents visit and use these spaces, thereby significantly enhancing their happiness experiences (*L.* Kong et al., 2022). In addition, vegetation landscapes are a defining feature of waterfront parks. Floral displays and lush greenery are associated with richer sensory experiences and higher perceived environmental quality, which may contribute to psychological restoration (Zhou et al., 2023). Such environments also serve as popular venues for social interactions, including parent–child activities and neighborhood gatherings (Bao et al., 2022; Veitch et al., 2022).

In contrast, waterfront greenways exhibit the lowest overall level of HII, the greatest annual fluctuation, and the strongest dependence on seasonal variation. SWB in greenways is most strongly associated with health-oriented physical activity, which shows the highest relative importance among the three typologies. Cycling and walking showed the highest relative importance among all activity-related indicators, reflecting the strong activity-oriented characteristics of waterfront greenways (He et al., 2021). Their linear layout and strong connectivity create ideal conditions for sustained physical activity, enabling residents to gain both physical vitality and psychological relaxation while staying close to nature (Xie et al., 2021). Moreover, favorable environmental conditions—such as good water quality, cleanliness, and fresh air—act as enabling factors for physical activity, contributing to greater comfort and enjoyment during exercise, as well as higher levels of happiness during such experiences. However, greenways also present notable negative factors, particularly the sense of environmental danger, which significantly suppresses SWB.

From a theoretical perspective, the observed differences in SWB across waterfront space types can be interpreted as the combined effects of Person–Environment Fit, restorative, and environmental load mechanisms. First, different waterfront space types offer heterogeneous behavioral opportunities, which in turn generate distinct activity structures. These behavioral differences can be explained through the Person–Environment Fit framework, emphasizing the match between individual needs and environmental affordances. Furthermore, variations in activity structures are likely to shape how well-being responses in different waterfront environments vary under changing climatic conditions. Previous studies have indicated that, compared with leisure and social activities, physical activity is generally more sensitive to climatic conditions (Ferguson et al., 2023; Turrisi et al., 2021). Accordingly, waterfront greenways, where walking and cycling dominate, exhibit the most pronounced seasonal variation. In contrast, waterfront parks, characterized by more static recreational use and higher vegetation coverage, demonstrate lower climate sensitivity and more stable levels of SWB. Secondly, from the perspective of Attention Restoration Theory, riverfronts, with their open views and diverse activity opportunities, are more likely to evoke key restorative environment characteristics, including being away, fascination, extent, and compatibility (Kaplan and Kaplan, 1989). These qualities in turn facilitate positive emotions and psychological restoration, leading to higher levels of SWB. Finally, from the perspective of Environmental Load Theory, this phenomenon provides a useful framework for understanding how spatial crowding and environmental risks shape SWB. Waterfront parks, due to their relatively limited spatial scale, are more susceptible to environmental load induced by crowding, which partially explains why their overall well-being levels are slightly lower than those of riverfronts (M. Lin et al., 2023; J. Zhang et al., 2023). In addition, waterfront greenways are often characterized by narrow linear configurations, with certain sections experiencing mixed pedestrian–vehicle flows, insufficient lighting, and limited visibility. These environmental conditions are more likely to generate feelings of insecurity and perceived environmental risk (Kwiatkowski and Karbowinski, 2023). Elevated perceptions of environmental risk may increase individual vigilance and psychological stress, thereby weakening positive emotional experiences and restorative effects. This further explains why

感中重要作用的研究结果不同（F.Liu et al.，2022；J.Zhang et al.，2025）。这一差异可能源于社交媒体文本数据的表征特性：相较于视觉景观与活动相关体验，声景体验更具瞬时性、情境依赖性，且在文本叙述中鲜有明确表达，因而相关体验在用户生成内容中的代表性可能不足。

进一步分析揭示了滨水特征与 SWB 之间存在显著的交互作用模式。某些特征表现出正向协同效应——例如，桥梁与步行在对 SWB 的贡献方面观察到正向交互作用。桥梁不仅提升了环境的视觉吸引力，还提供了延伸的步道和视觉焦点，从而丰富了步行体验（Plasencia-Lozano，2015）。相比之下，公共交通对 SWB 的贡献则与空气新鲜度呈负向交互作用，这可能是由于频繁的交通流量及其伴随的排放物对人们主观感知空气质量产生了负面影响。这些研究结果表明，SWB 与多种特征相关，而非仅与单一因素相关，且这些特征之间的关系在特定情境下会发生变化。

5.2. SWB 不同滨水类型间的差异与关联模式

不同类型的滨水空间在健康影响指数（HII）水平及时间动态特征方面均存在显著差异，这些差异与其体验特性的差异密切相关。既往研究通常依据水体属性（如河流、湖泊与运河）对滨水空间进行分类，并比较不同水体类型与健康相关结局之间的关联（Gao与Fang，2025；F.Liu等，2022；Luo等，2023；McDougall等，2022）。本研究通过拓展滨水空间与健康关系的分类范式及理论阐释，对现有研究进行了延伸。具体而言，本研究不仅揭示了不同滨水空间类型在幸福感水平及季节性动态特征上的差异，更进一步探究了这些差异背后的潜在成因。

具体而言，滨水区域展现出最高的整体HII水平，并伴有适度的季节性波动，这与多样化的行为结构——尤其是体力活动与休闲娱乐的结合——密切相关，而这种行为结构又与积极情绪及心理恢复相关联。宽阔且连续的滨水步道，以及无障碍的空间布局，与体力活动、更高水平的积极情绪及心理活力显著相关（Yan 等，2024）。与此同时，诸如享受河风或欣赏风景等休闲活动，营造出典型的情绪恢复与感官愉悦场景，使游客通过视觉与听觉刺激得以放松身心，缓解精神疲劳与认知负荷（Severin等，2022；G.Zhu等，2023）。滨水区域的建筑与景观特征——尤其是标志性的长江大桥——连同开阔的河景及周边的滨水建筑，共同构成了层次丰富的视觉界面。这种景观布局不仅通过提升空间的视觉吸引力来丰富锻炼体验，还通过强化城市形象与地域特色，增强了滨水区的休闲吸引力（Lynch，2023）。

滨水公园的整体HII水平略低于河岸公园，但全年仍保持最稳定的状态，对季节及气候变化的敏感性也最为弱。在滨水公园中，交通可达性与环境质量均与 SWB 呈现显著联合关联。相较于河岸公园与绿道，滨水公园通常以微观尺度嵌入住宅区，且空间分布较为分散，因而成为居民日常访问中最常用的滨水环境。因此，交通感知是滨水公园中与 SWB 关联最强的因素。交通可达性越高，居民访问并使用这些空间的频率与强度就越大，从而显著提升其幸福感体验（*L*.Kong等，2022）。此外，植被景观是滨水公园的标志性特征。繁茂的植被与丰富的花卉景观，可带来更丰富的感官体验及更高的环境感知质量，进而有助于实现心理修复（Zhou等，2023)。此类环境也成为社交互动的热门场所，包括亲子活动与社区聚会（Bao 等，2022；Veitch 等，2022）。

相比之下，滨水绿道在HII总体水平上最低，年度波动幅度最大，且对季节变化的依赖性最强。绿道中的 SWB 与健康导向型体力活动关联最为密切，这一指标在三种类型中显示出最高的相对重要性。骑行与步行在所有与活动相关的指标中均表现出最高的相对重要性，这反映了滨水绿道强烈的活动导向特性（He 等，2021）。其线性布局与强大的连通性为持续体力活动创造了理想条件，使居民在亲近自然的同时，既能提升身体活力，又能获得心理放松（Xie 等，2021）。此外，良好的环境条件——如优质的水质、洁净的环境及清新的空气——也为体力活动提供了有利条件，不仅提升了运动过程中的舒适度与愉悦感，还增强了此类体验中的幸福感。然而，绿道也存在着显著的负面因素，尤其是环境危险感，这一因素会显著抑制 SWB 。

从理论角度来看，不同滨水空间类型中观察到的 SWB 差异，可解释为“人-环境适配”、“恢复性”及“环境负荷”机制共同作用的结果。首先，不同的滨水空间类型提供了多样化的行为机会，进而形成了不同的活动结构。这些行为差异可通过“人-环境适配”框架来解释，该框架强调个体需求与环境提供条件之间的匹配度。此外，活动结构的差异也可能影响不同滨水环境中幸福感反应在气候变化条件下的变化程度。既往研究表明，与休闲及社交活动相比，体力活动对气候条件的敏感性通常更高（Ferguson 等，2023；Turrisi 等，2021）。因此，以步行和骑行为主的滨水绿道，其幸福感反应呈现出最为显著的季节性变化。相比之下，滨水公园以较为静态的休闲用途和较高的植被覆盖率为特征，表现出较低的气候敏感性和更为稳定的 SWB 水平。其次，从注意力恢复理论的视角来看，滨河区凭借其开阔的视野和丰富的活动机会，更有可能激发关键的恢复性环境特征，包括远离喧嚣、引人入胜、空间广阔以及环境适宜性（Kaplan 与 Kaplan，1989）。这些特性进而促进积极情绪和心理恢复，从而提高 SWB 水平。最后，从环境负荷理论的视角出发，这一现象为理解空间拥挤与环境风险如何影响 SWB 提供了有用的框架。由于滨水公园的空间规模相对有限，它们更容易受到由拥挤引发的环境负荷影响，这部分解释了为何其整体幸福感略低于滨河区（M.Lin等，2023；J.Zhang等，2023）。此外，滨水绿道通常具有狭窄的线性布局特征，其中某些路段会出现行人与车辆混杂流动、照明不足以及可视性有限等问题。这些环境条件更容易引发不安全感及感知到的环境风险（Kwiatkowski 与 Karbowinski，2023）。对环境风险的更高感知可能会增强个体的警觉性并加剧心理压力，从而削弱积极情绪体验与恢复效

greenways exhibit lower overall SWB compared with riverfronts and waterfront parks. This finding further extends the analytical scale of blue-space research.

5.3. Implications for theory and practice

This study broadens the analytical scope regarding the linkage between urban waterfront spaces and SWB, providing novel empirical evidence for understanding heterogeneous health benefits across diverse waterfront environments. Theoretically, this study advances existing waterfront classification approaches by shifting the conventional waterbody-based categorization toward a framework grounded in spatial morphology and functional attributes. It systematically classifies urban waterfront spaces into three representative types: riverfronts, waterfront parks, and waterfront greenways, and further reveals the differences among these spatial types in terms of SWB association patterns and seasonal dynamics. Furthermore, by integrating person–environment fit theory, restorative environment theory, and environmental load theory, this study provides theoretical interpretations of the differentiated relationships between waterfront types and SWB. Beyond previous studies that have mainly focused on objective built environment attributes, this research shifts attention toward human-centered experiential dimensions by developing a refined waterfront experience indicator framework encompassing environmental perception, environmental cognition, and behavioral activities. The framework enables a systematic examination of the associations between multidimensional experiential factors and SWB. The findings suggest that the health benefits of waterfront environments are not solely determined by physical environmental characteristics but are closely associated with how individuals perceive, evaluate, and interact with these environments. This experience-based typological framework enriches existing blue space health research and provides new empirical insights into the differentiated well-being benefits of diverse waterfront spaces.

Practically, the findings provide actionable and health-oriented guidance for typology-specific waterfront planning and policy-making. First, physical activity emerges as the primary driver of waterfront SWB. Spatial design should therefore prioritize continuous and high-quality walking, running, and cycling networks, alongside improvements in lighting, safety, and resting facilities to enhance the health-promoting potential of waterfront environments. Meanwhile, Traffic perception and perceived crowding represent fundamental conditions for positive user experiences. Enhancing transport connectivity, optimizing pedestrian flow, and improving spatial carrying capacity can effectively strengthen accessibility and reduce negative perceptions of overcrowding. Second, differentiated strategies are required across waterfront typologies. Riverfronts should leverage open spatial structures and iconic landscapes to integrate physical activity with scenic appreciation. Waterfront parks should strengthen accessibility, recreational amenities, and cultural narratives to foster socially and emotionally engaging environments. Waterfront greenways, by contrast, should maintain their activity-oriented functions while prioritizing safety and environmental quality—such as improved lighting, monitoring, and water management—to address their higher sensitivity to environmental constraints. Finally, waterfront SWB reflects the combined effects of multiple environmental and behavioral factors. Planning should therefore move beyond single-factor optimization toward integrated and adaptive strategies, balancing key elements and recognizing potential threshold effects. From a policy perspective, both equity and differentiation are essential: ensuring equitable access to basic activity and leisure opportunities while reinforcing the distinctive functions of different waterfront types. This approach supports more efficient resource allocation, enhances public happiness, and promotes healthier and more sustainable urban development.

5.4. Limitations and future work

Although this study provides an innovative perspective and methodological framework for analyzing the relationship between waterfront spaces and SWB, several limitations remain. From a data perspective, first, there is a potential sample selection bias. Active users on the Dianping platform are predominantly young and middle-aged urban residents with relatively high digital literacy, while older adults and minors are underrepresented in online expressions (Ghermandi and Sinclair, 2019; Su and Sun, 2025). As a result, the dataset does not fully capture all age groups and social strata, which may systematically underestimate the health needs and experiential perceptions of vulnerable populations regarding waterfront spaces. Second, there are limitations in data dimensionality. Due to platform privacy regulations and restrictions on data access, this study could not obtain key sociodemographic variables such as age, gender, income level, and educational background. This constraint prevents the conduct of heterogeneity analyses and limits the ability to reveal differentiated patterns of SWB across social groups. Consequently, the effects of certain experiential factors on health outcomes may be misestimated, thereby constraining the external validity and generalizability of the findings (Huang et al., 2025).

From the perspective of measurement, the assessment of SWB in this study is relatively unidimensional. This study primarily relies on text-based sentiment analysis to capture public emotional expressions, focusing on immediate affective experiences, while failing to fully encompass the multidimensional nature of health effects associated with waterfront spaces. In fact, the health benefits of waterfront environments extend beyond momentary emotions to include life satisfaction, psychological and physiological restoration, and social well-being. Sole reliance on textual sentiment data therefore cannot adequately represent the comprehensive health value of waterfront spaces (Diener, 1984).

From a methodological perspective, this study primarily relies on social media text data to examine human–environment interactions, which inevitably imposes certain limitations on both analytical scope and causal inference. While textual data can effectively capture users' subjective experiences and evaluations, they are unable to comprehensively represent objective environmental exposure characteristics of waterfront spaces, such as visual landscape attributes, activity settings, and other physical environmental conditions. Furthermore, the findings of this study should be interpreted as statistical associations rather than causal effects. Although the SHAP-based explainable machine learning framework can effectively identify the relative contribution of experiential factors and reveal complex nonlinear association patterns, the cross-sectional nature of social media data does not allow causal relationships between waterfront experiences and SWB to be established.

Future research could be further advanced in three directions. First, mixed-methods approaches combining field surveys and individual-level tracking studies could be employed to address the limitations of social media data in terms of sample representativeness and individual information. Specifically, targeted investigations involving vulnerable groups, such as older adults and children, should be conducted, while collecting detailed individual characteristics, including sociodemographic attributes, visit frequency, duration of stay, and travel modes, to better reveal heterogeneous waterfront health experiences across different populations. Second, future studies could establish a multidimensional framework for assessing health effects by moving beyond single-dimensional emotional measurements. Psychological scales such as the WHO-5, GHQ-12, and perceived restoration measures, together with wearable health monitoring devices and health records, could be integrated to comprehensively evaluate the psychological, physiological, and social benefits of waterfront environments. Third, future research could incorporate multimodal big-data approaches by integrating image-based analysis to extract environmental characteristics, such as vegetation coverage, sky openness, and visual complexity, with

应。这进一步解释了为何绿道的整体 SWB 值低于河岸带及滨水公园。这一发现拓展了蓝空间研究的分析维度。

5.3. 对理论与实践的启示

本研究拓展了关于城市滨水空间与 SWB 之间关联性的分析范畴，为理解不同滨水环境中多样化的健康效益提供了新的实证依据。在理论层面，本研究通过将传统的基于水体的分类方法转向以空间形态和功能属性为基础的框架，推动了现有滨水分类方法的发展。研究系统地将城市滨水空间划分为三种代表性类型：河岸带、滨水公园和滨水绿道，并进一步揭示了这些空间类型在 SWB 关联模式及季节性动态方面的差异。此外，本研究整合了“人-环境适配理论”、“恢复性环境理论”及“环境负荷理论”，为不同滨水类型与 SWB 之间的差异化关系提供了理论阐释。相较于以往主要关注客观建成环境属性的研究，本研究通过将一个精炼的水岸体验指标框架——涵盖环境感知、环境认知及行为活动——加以开发，将研究焦点转向以人为中心的体验维度。该框架有助于系统性地考察多维体验因素与 SWB 之间的关联。研究结果表明，水岸环境对健康的益处并非仅由物理环境特征决定，而是与个体对这些环境的感知、评估及互动方式密切相关。这一基于体验的分类框架丰富了现有蓝空间健康研究，并为不同水岸空间带来的差异化福祉效益提供了新的实证见解。

实际上，这些研究结果为特定类型的滨水区规划与政策制定提供了可操作且以健康为导向的指导。首先，体力活动成为滨水区 SWB 的主要驱动力。因此，空间设计应优先考虑连续且高质量的步行、跑步及骑行网络，同时改善照明、安全性和休息设施，以增强滨水环境的健康促进潜力。与此同时，交通感知与感知拥挤度是打造积极用户体验的基础条件。提升交通连通性、优化行人流量以及提高空间承载能力，可有效增强可达性，减少对过度拥挤的负面感知。其次，针对不同类型的滨水区需采取差异化策略。河岸区应利用开放式空间结构和标志性景观，将体力活动与景观欣赏相结合。滨水公园则应加强可达性、休闲设施及文化叙事，以营造兼具社会与情感吸引力的环境。相比之下，滨水绿道应在保持其以活动为导向的功能的同时，优先考虑安全性和环境质量——例如改善照明、监测及水管理——以应对其对环境约束的更高敏感性。最后，滨水 SWB 反映了多种环境及行为因素的综合作用。因此，规划工作应超越单一因素优化，转向综合且适应性的策略，平衡关键要素，并识别潜在的临界效应。从政策角度来看，公平性与差异化均至关重要：既要确保公众公平享有基本的活动与休闲机会，又要强化不同滨水类型的独特功能。这一方法有助于更高效地配置资源，提升公众幸福感，并促进更健康、更可持续的城市发展。

5.4. 局限性与未来研究方向

尽管本研究为分析滨水空间与 SWB 之间的关系提供了创新视角和方法学框架，但仍存在若干局限性。从数据角度来看，首先，可能存在样本选择偏差。Dianping平台上的活跃用户主要为年轻及中年城市居民，其数字素养相对较高，而老年人及未成年人在线表达的比例较低（Ghermandi 与Sinclair，2019；Su 与 Sun，2025）。因此，该数据集未能全面涵盖所有年龄层和社会阶层，这可能导致对弱势群体在滨水空间方面的健康需求及体验感知被系统性低估。其次，数据维度也存在局限。由于平台隐私法规及数据访问限制，本研究无法获取年龄、性别、收入水平及教育背景等关键社会人口学变量。这一限制阻碍了异质性分析的实施，也限制了揭示不同社会群体间 SWB 差异模式的能力。因此，某些经验性因素对健康结局的影响可能被误估，从而限制研究结果的外部效度与普适性（Huang 等，2025）。

从测量的角度来看，本研究中对 SWB 的评估相对单一。本研究主要依赖基于文本的情感分析来捕捉公众的情感表达，侧重于即时情感体验，却未能全面涵盖滨水空间相关的健康效应的多维特性。事实上，滨水环境的健康效益不仅限于瞬时情感，还包括生活满意度、心理与生理的恢复以及社会福祉。因此，仅依赖文本情感数据无法充分反映滨水空间的综合健康价值（Diener，1984）。

从方法论的角度来看，本研究主要依赖社交媒体文本数据来探讨人与环境的互动，这不可避免地对分析范围和因果推断施加了某些限制。尽管文本数据能够有效捕捉用户的主观体验和评价，但它们无法全面反映滨水空间的客观环境暴露特征，如视觉景观属性、活动环境及其他物理环境条件。此外，本研究的结果应被解读为统计关联，而非因果效应。尽管基于Shap值的可解释机器学习框架能够有效识别体验因素的相对贡献，并揭示复杂的非线性关联模式，但社交媒体数据的横断面特性仍无法确立滨水体验与 SWB 之间的因果关系。

未来研究可从以下三个方向进一步深化：首先，可采用结合实地调查与个体层面追踪研究的混合研究方法，以克服社交媒体数据在样本代表性及个体信息维度上的局限性。具体而言，应针对老年人及儿童等易感人群开展定向研究，同时采集详尽的个体特征数据——包括社会人口学属性、到访频率、停留时长及出行方式——从而更精准地揭示不同人群在水滨环境中的健康体验差异。其次，未来研究可构建多维健康效应评估框架，突破单一维度情绪测量的局限。可将WHO-5量表、GHQ-12量表及感知恢复量表等心理测量工具，与可穿戴健康监测设备及健康记录相结合，系统评估滨水环境对心理、生理及社会层面的综合效益。第三，未来研究可引入多模态大数据分析方法，将基于图像的分析技术（用于提取植被覆盖率、天空开阔度及视觉复杂度等环境特征）与时空行为数

spatiotemporal behavioral data, including mobile positioning trajectories and stay duration. Such data fusion could provide deeper insights into the mechanisms through which waterfront environments influence public health and well-being, while offering more robust evidence-based support for urban public space planning and policymaking.

6. Conclusion

This study integrates large-scale social media data, deep learning, and explainable machine learning approaches to investigate the association patterns between SWB and public experiences across different types of urban waterfront spaces. By developing a comprehensive analytical framework encompassing perceptual, cognitive, and behavioral dimensions, the study provides a novel perspective for understanding the interactions between human health and urban environments. The results indicate that SWB in waterfront spaces is closely associated with multiple experiential factors, including behavioral, perceptual, and cognitive experiences, among which behavioral activities play a dominant role. Moreover, distinct well-being characteristics and association patterns were observed across different waterfront typologies, suggesting that the relationships between experiential factors and SWB are highly context-dependent. These findings highlight the importance of examining typological differences among waterfront spaces and provide valuable empirical evidence for the optimization of urban waterfront environments and health-oriented design practices.

Declaration of generative AI in scientific writing

Statement: During the preparation of this work the authors used GPT-based language model in order to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Ke Wang: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation, Conceptualization. **Xia Zhang:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Hong Yang:** Writing – review & editing, Supervision, Investigation. **Luliang Tang:** Writing – review & editing, Supervision, Funding acquisition. **Dong Xie:** Writing – review & editing, Validation, Conceptualization. **Cunyu Yuan:** Writing – review & editing, Validation, Data curation.

Ethical approval

This study was approved by the Ethics Committee of Wuhan University (Approval No. WHU-NS-IRB2025001). The research used publicly available social media data from Dianping. All data were anonymized during processing, and no personally identifiable information was collected or analyzed.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ecolind.2026.115367.

Data availability

Data will be made available on request.

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据（包括移动定位轨迹及停留时长）相结合。此类数据融合技术不仅能深入揭示滨水环境影响公众健康与福祉的作用机制，还可为城市公共空间规划及政策制定提供更坚实的循证支持。

6. 结论

本研究整合了大规模社交媒体数据、深度学习及可解释机器学习方法，旨在探究 SWB 与不同类型城市滨水空间中的公众体验之间的关联模式。通过构建一个涵盖感知、认知及行为维度的综合分析框架，本研究为理解人类健康与城市环境之间的相互作用提供了新的视角。研究结果表明，滨水空间中的 SWB 与多种体验因素密切相关，包括行为、感知及认知体验，其中行为活动起着主导作用。此外，在不同类型的滨水空间中，还观察到了不同的幸福感特征及关联模式，这表明体验因素与 SWB 之间的关系高度依赖于具体情境。这些发现强调了考察滨水空间类型差异的重要性，并为优化城市滨水环境及健康导向设计实践提供了宝贵的实证依据。

科学写作中生成式人工智能声明

声明：在本研究撰写过程中，作者采用了基于 GPT 的语言模型，以提升论文的可读性与语言质量。在使用该工具后，作者根据需要对内容进行了审阅与修订，并对已发表文章的内容承担全部责任。

CRediT 作者贡献声明

Ke Wang：文稿审阅与编辑、文稿初稿撰写、可视化、软件支持、研究方法、调研、数据管理、概念化。**Xia Zhang：**文稿审阅与编辑、项目监督、调研、资金获取、形式分析、概念化。**Hong Yang：**文稿审阅与编辑、项目监督、调研。**Luliang Tang：**文稿审阅与编辑、项目监督、资金获取。**Dong Xie：**文稿审阅与编辑、验证、概念化。**Cunyu Yuan：**文稿审阅与编辑、验证、数据管理。

伦理批准

本研究已获得武汉大学伦理委员会批准（批准号： WHU -NS-IRB 2025001）。研究使用了来自大众点评平台的公开社交媒体数据。所有数据在处理过程中均经过匿名化处理，未收集或分析任何个人可识别信息。

利益冲突声明

作者声明，其不存在已知的竞争性经济利益或个人关系，这些利益或关系可能对本研究结果产生不良影响。

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附录 A：补充数据

本文的补充数据可在以下网址在线获取：https://doi.org/10.1016/j.ecolind.2026.115367。

数据可用性

相关数据可根据要求提供。

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